

Global Facilities - Design & Construction Standard - Chemical - Chemical Blender System Specification

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1.0 Purpose

This Standard Specification defines the minimum requirements for design, engineering, fabrication, installation, supply, test & commission, certify, validate the performance, train the Owner's personal in the operation and maintenance of systems, and guarantee the pressure/flow/purity/safety performance for chemical blender system. This document is part of Micron's general specification standards developed to meet the process requirements while requirements for safety is duly considered.

TD may decide to have a scaled down version of the chemical blending system but shall generally follow the design concepts where possible as the consumption and capacity requirement is much smaller and downtime impact is not a major concern as compared to a HVM sites.

Note:

1. If any sites practice (or require) more stringent standards / specifications / regulations on compliance/etc., those sites may have to follow their existing (more stringent) standards/ specifications/ regulations on compliance/ etc. in order to meet site specific requirements, standards and specifications.
2. [Global Facilities - Design & Construction Standard - Chemical - Chemical Distribution System](#) shall be used for the Blended Chemical Distribution System that includes drum/tote/IBC module enclosure, chemical transfer units, chemical dispense module and other associated modules such as day tanks, distribution piping network, VMBs, MHB(T-Box), MHEB (expansion box), JB etc. Works related to Electrical, Instrumentation and Controls, and Mechanical ducts and air exchange rates shall be referred to Chemical distribution system design standard or the respective standards based on the

reference stated in chemical distribution system design standard. As process tools are sensitive to required pressure, flow and quality, dispense modules are required to operate within the parameter specifications.

2.0 Scope

2.1 Work Scope:

- a) Design and installation of Chemical blending systems in any chemical rooms that are in specified area. Based on requirements, drawings for location of the rooms will be provided.
- b) This standard covers both in-line blender and batch blender.
- c) This standard will not address slurry blending systems, chemical dispense systems or distribution of the blended material from dispense system
- d) Interconnecting piping/tubing between chemical blending system and VMB or other Distribution systems with required secondary containment.
- e) Hook-up lines from other utilities such as UPW, PCW, ICA, CDA, PN2, UN2, drain, etc. to different system modules and other required components.
- f) Blending system installation based on the system installation room criteria.
- g) Control, Instrumentation, dedicated monitoring system SCADA (based on requirement) and Data concentrator. If there is existing control system network, required additions/modifications shall be made.
- h) FAT, SAT, installation inspection, testing, start-up, commissioning, validation and training

2.2 General Scope

This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects. However, based on requirements and applicability with existing design, this document can also be used for Brownfield and other small expansion projects.

Note: This document does not apply to organic chemicals and deposition precursors such as TEOS, TEB, TEPO, HCDS, TiCl₄, and all other products that are supplied by dedicated cabinets and are normally associated with the gas systems.

3.0 Definitions and Acronyms

- **APV** – New PTFE APV (**A**dvanced **P**erformance **V**essel), aka Modified PTFE
- **ATEX** – **EX**plosive **At**mospheres
- **AOV** – Air Operated Valve
- **BKM** – Best Known Method
- **CDA** – Clean Dry Air.
- **CFD** – Computational Fluid Dynamics
- **CTU** – Chemical Transfer Unit
- **CTDM** – Chemical Transfer Dispense Module
- **CDM** – Chemical Dispense Module
- **CO2** – Carbon Dioxide
- **COHE** – Control of Hazardous Energy
- **ΔP** – Delta Pressure
- **EPCM** - Engineering Procurement and Construction Management
- **ET** - Etch: Chemical etched sheet on the back of lining material for bonding to shell
- **ERT** – Emergency Response Team
- **EHS** – Environmental, Health & Safety
- **FAT** – Factory Acceptance Test
- **FRP** – Fiber-reinforced plastic
- **FEP** - Fluorinated Ethylene Propylene
- **FCT** – Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
- **FIOP** – Field Input Output Panel
- **FM** – Factory Mutual
- **GN2** – General Nitrogen
- **GB** – Glass Backing, intermediary sheet bonded between tank liner and shell
- **GFTT** – Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
- **GFC** – Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities

- **Global Facilities Central Team** – This is used interchangeably with Innovation & Integration Team
- **Global Facilities Construction Team** – This is used interchangeably with Global Construction Team
- **Global Facilities Planning Team** – This is used interchangeably with Global Facilities Line Planning Team
- **Global Facilities Operations Support Team** – This is used interchangeably with Global Facilities Operations Team
- **HAZOP** – Hazardous Operation
- **HAZMAT** – Hazardous Material
- **HMI** – Human Machine Interface
- **HDPE** – High Density Polyethylene
- **HCDS** – Hexachlorodisilane
- **HVM** – High Volume Manufacturing
- **I&C** – Instrumentation and Control
- **ICA** – Instrument Control Air
- **JB** – Junction Box with no valves
- **LSS** – Life Safety System
- **LPP** – **L**ow **P**ermeation **P**roperties, two-layer PFA tube
- **MHB** – Main Header Box
- **MHEB** – Main Header End Box
- **MTBF** – Mean Time Between Failure
- **MTTR** – Mean Time To Repair
- **MV** – Manual Valve
- **N2** – Nitrogen
- **NPW** – Non-Portable Water
- **NFPA** – National Fire Protection Association
- **PCW** – Process Cooling Water
- **PLC** – Programmable Logic Controller
- **PN2** – Process Nitrogen
- **PTFE** – Polytetrafluoroethylene
- **PLP** – Purified Low Permeation, aka Modified PTFE

- **PTFM** – Modified PTFE
- **P&ID** – Plumbing and Instrumentation Diagram
- **P&FM** – Petroleum and Flammable Materials
- **PSM** – Process Safety Management
- **PFA** - Perfluoroalkoxy Alkane
- **PP**- Polypropylene
- **PVC**- Polyvinyl Chloride
- **PVDF** - Polyvinylidene Difluoride
- **RFP** – Request for Proposal
- **RRA** – Risk Ranking Analysis
- **RIO** – Remote Input / Output Panel
- **RPM** – Revolution Per Minute
- **R & R** – Repeatability and Reproducibility
- **RE** – Relative Error
- **SAT** – Site Acceptance Test
- **SDS** – Safety Data Sheet
- **SPOF** – Single Point of Failure
- **SS** – Stainless Steel
- **SOO** – Sequence of Operation
- **TD** – Technology Development
- **TGCM** – Total Gas & Chemical Management
- **TMAH** – Tetramethyl Ammonium Hydroxide
- **TEOS** - Tetraethyl Orthosilicate
- **TEB** – Triethyl Borane
- **TEPO** – Triethyl Phosphate
- **TiCl₄** - Titanium Tetrachloride
- **UN₂** – Utility Nitrogen
- **UV/IR** – Ultraviolet/Infrared Ray
- **UPW** – Ultra Pure Water
- **UHP** – Ultra High Purity
- **UOM** – Unit of Measure
- **VMB** – Valve Manifold Box
- **VFD** – Variable Frequency Drive

4.0 Definitions and Acronyms

Roles	Responsibilities
GFTT Lead	• Develop and maintain System Standard document • Incorporate Best Known Methods (BKM), lessons learned, and emerging technologies into the System Standard • Perform periodic document review and update all relevant components in System Standard • Support the Site teams by providing knowledge and technical expertise
FCT Lead	• Review and provide technical feedback on System Standard • Update BKMs and Develop relevant BKMs • Support the Site teams by providing knowledge and technical expertise in operation and maintenance
Site Project Construction Team	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard • Consult with Global/Site PSM on the Relative Ranking Analysis (RRA) and ensure the RRA is less than 30. Otherwise carry out mitigation till RRA comes below 30.
Site Facilities Team	• Use System Standard as technical reference for new system designs
System Supplier or other contractors	• Comply or exceed the requirements mentioned in relevant Micron standards
Global/Site PSM	• Provide guidance for Relative Ranking Analysis rating for systems • Provide consultancy on the required mitigation

5.0 Safety

5.1 Project Safety Requirements

- a) The EPCM contractor or system vendor's written EHS program shall contain all the necessary elements to administer the program in accordance with all applicable EHS Laws, Micron corporate and site-specific safety policies as well as local regulatory requirements. Compliance with the below mentioned link to Micron safety standards shall also be adhered.
- i. [Global EHS - Construction Contractor EHS Minimum Performance Requirements](#)
 - ii. [Global EHS - Construction EHS Management Strategy](#)
 - iii. [Global EHS - Contractor and Vendor EHS Standard](#)
 - iv. [Global EHS - Contractor EHS Requirements](#)
 - v. [Global EHS - Cranes and Rigging Standard](#)
 - vi. [Global EHS - Confined Space Program Standard](#)
 - vii. [Global EHS - Control of Hazardous Energy \(CoHE\) Standard](#)
 - viii. [Global EHS - Electrical Safety Standard](#)
 - ix. [Global EHS - Work At Heights Standard](#)
 - x. [Global EHS - Lone Worker Standard](#)
 - xi. [Global EHS - Standard for Storing Flammables and Combustibles](#)
 - xii. [Global EHS - TMAH Safety Standard](#)
 - xiii. [Global EHS - Electrical Safety Standard](#)
 - xiv. [Global EHS - Safety Permit To Work Standard](#)
- b) Contractors / system vendors and Micron site project team are expected to assess all related safety risk factors during chemical receiving, handling, flushing, commissioning and sampling activities. Preventive and contingency plans shall be created to address all safety concerns (Job Hazard Analysis (JHA) or Risk Assessments shall be conducted as needed).
- c) Personal Protective Equipment (PPE) – All contractor's /system vendor's personnel shall adhere to their own mandated PPE requirements for chemical handling and shall be reviewed with Micron site's Environmental Health, and Safety, (EHS) for adequacy.
- d) Some examples of major safety concerns include, but are not limited to:
- General area safety (e.g. tripping hazards)
 - Potential chemical exposure / leaks / spill
 - Waste management (e.g. Hazardous)
- e) Contractors / system vendors are expected to ensure that their Team Members and sub-contractors involved in chemical handling activities are adequately trained (with valid certification if applicable) for conducting the task. This should include all necessary in-house and specialized training specific to equipment or system (if applicable).

- f) Chemical Information - Chemicals encountered during execution of commissioning procedure may vary as such refer to chemical information labels affixed to the equipment or specific tool and the Safety Data Sheet (SDS).
- g) Descriptions of Danger, Warning, Caution, Notice, and Important.
 - **Danger:** Indicates a hazardous situation that, if not avoided, will result in death or a permanent disabling injury.
 - **Warning:** Indicates a hazardous situation which, if not avoided, could result in death or severe injury.
 - **Caution:** Indicates a hazardous situation that, if not avoided, may result in moderate or minor injury.
 - **Notice:** Indicates a hazardous situation that, if not avoided, may result in property damage.
 - **Important:** Alerts the user to information that is crucial to a step.

6.0 General Requirements and Design Concept

6.1 General Requirements:

- a) The chemical blending System shall be provided, installed, and commissioned by selected Contractor. The Contractor shall have sole responsibility for compliance with the project Contract Documents, and shall coordinate with other sub-contractors, the Owner and/or EPCM to complete the System according to the project schedule.
- b) Complete system includes but not limited to delivery and installation of all required materials, components, auxiliaries, fittings, fastenings, supports, labelling, training of the operation personnel; temporary provisions; vertical/horizontal transportation, utilities for commissioning like temporary hoses for drains, etc. This means that everything is included from design to fabrication to installation to testing & commissioning to hand over in obtaining correct and perfect operating systems, even if such items are not specifically mentioned in the specifications or indicated in the drawings.
- c) The System shall be designed, fabricated, installed, and validated in compliance with the requirements stated in this document and associated standards.
- d) As chemical Blending systems are supposed to operate with hazardous (toxic, corrosive and flammable) media, the required code and local regulation compliance shall be strictly observed in the design and installation operation of the systems can be carried out without any non-compliance.
- e) Review the make-up air and room exhaust system design, including distribution ducting, inside the designated areas. For some chemical

rooms, the make-up air room main outlet and exhaust inlet ducts have to be installed down to 50cm above the floor. Assure that these locations are coordinated.

- f) The System shall be suitable for continuous operation and therefore, it shall be designed with minimal maintenance and operator intervention. The facility shall consistently supply blended Chemical quality within specification and shall operate 365 days per year, 7 days per week, 24 hours per day, with no unscheduled down time.
- g) Unit quantities, system sizing, flow rates, pressures, temperatures, etc. stated in this standard (or specification related to chemistry will be provided during design review) are general guidelines. However, the contractor must make his own evaluation in order to achieve the required pressure, temperature, etc. at the specified quantity and quality for each point of use. Piping distribution drawings, equipment layout drawings, and design matrixes shall be provided to enable pressure and flow calculations. The contractor shall be liable for guaranteeing that the systems, including all sub-systems (if any), can meet all requirements within the RFP including required flow rates.
- h) All systems and related equipment near forklift paths to be protected from potential damage from forklift, pallet lifters, or payload impacts. Walls and wall corners, or anything protruding from walls that may be susceptible to damage should also be protected (e.g. steel plates, etc.).
- i) Depending on the country of system installation, all local code requirements shall be met and therefore contractor shall be responsible for all associated costs/fees to obtain the necessary permits/written approvals. In addition, all documents submissions/clearance to/from authority for written approvals for construction/installation/operation approvals where applicable shall be included. Including all necessary PE endorsements for Micron to obtain their P&FM license.
- j) Contractor shall be fully responsible for understanding what a local PE requires, when he reviews a sub-system that is a listed chemical in the P&FM list, in regards to assuring system integrity and/or compliance with the latest revision of the local Import of Petroleum & Flammable Materials (P&FM: Fire Safety Regulations).
- k) Work included consists of the furnishing of all labour, materials, tools, consumables. Transportation, lifting devices, services, permits, fees, utility connection fees, professional fees, taxes etc., which are applicable, required, and/or necessary for a complete and proper installation of the chemical blender systems as defined within this standard. Process chemicals will be provided by Micron in a quantity as proposed by the suppliers in their quotations.
- l) This standard represents the minimum requirements but does not include specific design details for the operational and performance requirements. It is the responsibility of the Contractor to provide a

complete and functional System and achieve proper system performance.

- m) Micron provides only typical P&ID (Refer **Appendix 8, 9, 10 & 11**) and the typical P&IDs are for illustration purpose only to show design concepts and so the actual P&ID may vary based chemical requirements, needs shall be finalized during design stage. Therefore, contractor needs to provide a complete design, including detailed P&ID's, process flow diagrams, 3D drawings/models, pipe and equipment plans and details, equipment skid general arrangements, storage vessel fabrication drawings, equipment support, pipe support and restraint details for Systems.
- n) Systems are to be designed based on the given specifications and contractor shall use either AFT Fathom 9.0 hydraulic or PIPE-FLO®, or an agreed software for modeling to validate flow requirement. Predicted Flow, Loop Pressure, and Pressure Drop across Filters, Valves, other components and along the loop shall be provided for approval before 3D modelling. 3D models for each system shall be provided for approval before construction.
- o) If other software such as CFD (Computational Fluid Dynamics), etc. to be used for blending, tank net volume calculations (Vortex and LL level setting), any mixer application, etc. Validated information shall be passed to Micron upon completion of software modelling.
- p) Contractor to submit source calculations to Micron to support design review.
- q) If Micron provides any drawings (such as P&ID, Layout, network architecture, single line diagrams, etc.) it is the scope of work of Contractor to take ownership of these drawings, update them as required, and provide additional drawings as needed following the project Legend Sheet requirements as well as standards for layering, symbology, and other criteria described in the project documentation standards and CAD standards.
- r) Relevant 3D drawings (or BIM) for rooms, Sub-Fabs and other areas shall be submitted so that these drawings will be superimposed to BIM to check the feasibility of installation without having conflicts with other utilities or equipment.
- s) Electronic copies of required documents shall be submitted by the contractor upon award.
- t) Space Management: Based on the drawings developed by the contractor to meet the requirements, contractor shall identify any additional and designated space requirements in the Bid.
- u) The Contractor shall consider repeated handling of the equipment, especially if he intends to store his equipment on-site. Storage of material/equipment up-to 6 weeks also forms part of the scope of the contractor. Prior to material/equipment move-in to site, Contractor shall provide a detail method statement & risk assessment on delivery and installation.

- v) The Contractor shall provide all necessary means for continuous protection of the equipment during construction, handling, testing, start-up, commissioning, and certification phases. Contractor shall provide fences/barriers and appropriate PPE during the qualification of the chemical blender systems. The contractor shall also provide a permanent PPE cabinet with appropriate PPE for chemical handling and spill control in each of the chemical room.
- w) The Contractor shall be responsible for the appropriate disposal of all waste generated during construction and commissioning. If mutual agreement between Micron and Contractor is not available with respect to cost on disposal of waste, the purchaser shall co-ordinate the disposal on a chargeable basis.
- x) If tender drawing design is not strictly followed, contractor is required to provide drawings to support alternative design in the bid and indicate deviation or exclusion.
- y) Any deviations from design and/or critical Micron requirements stated in this standard, either temporary or permanent shall be documented in the "Deviation Form" in **Appendix 7** and shall be approved prior to implementation.

6.1.1 System Robustness and Reliability Requirements

System robustness is a very critical factor in the design of these systems and therefore, the due importance shall be given by the contractors. As it is known, the Fab operates continuously and therefore, supply of consistent quality chemicals with consistent pressure & flow is required 24 hours per day, 365 days per year. Thus, contractors shall ensure the followings:

- a) To provide Mean Time Between Failure (MTBF) analysis data for equipment and components being proposed.
- b) To provide Mean Time to Repair (MTTR < 3 Hours) guarantee for both response and repair time, 1.5 hour + 1.5 hour (or as per site requirements).
- c) Operational requirement: capable of fully automatic operation; periodic maintenance and calibration activities without any impact to Fab operations. Optimize water, Instrument air, CDA, Nitrogen and energy conservation and minimize generation of chemical to waste streams.

Likewise, reliability and functionality of the system are also very critical to ensure operating under different circumstances. Thus, contractors shall ensure the followings:

- a) To provide control system including instrumentation, redundant or highly reliable PLC's, software, source code, operator interface terminals, supervisory control and data

acquisition (SCADA) software. Interface to and integrate with site facility management and control system (FMCS) and life safety system (LSS).

- b) To provide tubing, piping, conduit, heat trace (if required), insulation, wiring, exhaust ductwork, structural supports, equipment restraints, vibration isolation, electrical equipment, instrumentation and controls plus other components and materials needed to provide complete and fully functioning Systems.
- c) Thermal Movement: Design piping to accommodate thermal expansion and contraction i.e. no degradation shall be observed on the mechanical lifetime of the piping and connected equipment. Contractor shall consider in their design, additional mitigation features for H₂SO₄ and other sub-systems where there is the potential for exothermic reactions.
- d) Seismic Performance: Contractor to provide equipment anchoring in conformance to the requirements of the site specification and the applicable local code.
- e) N+1 redundancy shall be provided for all major system components that require routine servicing and /or repair. This includes Pumps, dampeners, etc.
- f) N+1 redundancy shall be provided for in-line blenders (whole blending line) and batch blenders
- g) Power distribution design to address electrical redundancy and single-point-of-failure mitigation. Systems shall be directly supplied from the Dual Feed AC Main Power Board having dual feed power supplies.

6.1.2 Performance Requirements

The Contractor shall be liable for the overall performance of the complete system as per contract, including guarantee on quality and flow. As mentioned in general requirements, the system shall be suitable for continuous operation, requiring minimal maintenance and operator intervention. The facility shall consistently supply blended Chemical quality within specification and shall operate 365 days per year, 7 days per week, 24 hours per day, with no unscheduled down time. Thus, performance requirements shall also include the followings:

- a) The Contractor shall be liable delivery and installation of all required materials, components, auxiliaries, fittings, fastenings, supports, labelling, training of the operation personnel; temporary provisions; vertical/horizontal transportation, utilities for commissioning like temporary hoses for drains, etc. This means that everything is included in obtaining correct and perfect operating systems, even if such items are not specifically mentioned in the specifications or indicated in the drawings, however essential for the correct operation.
- b) Provide a detailed qualification plan and milestone schedule within 30 days of the contract award. Qualification plan and sampling plan along with Micron and/or EPCM shall be finalized within 45 days of contract award.
- c) Site Acceptance Tests (SATs), Functional Acceptance Tests (FATs) and Factory Acceptance Test (FAT) shall be performed and documented. FATs shall include, but are not limited to, PLC interlock performance, HMI functionality, alarm functionality, and instrument and sequence of operation compliance. Evidence of test e.g. SCADA, HMI alarm, trend data shall be attached as a proof of testing for SAT. Any software bugs identified during system commissioning or after handover to Micron operations shall be rectified without any cost impact.
- d) The SCADA system shall be fully tested to be aligned to the system controller, and each limit and alarm documented. Analog values shall have a maximum deviation of 0.1 UOM or 1% of the intended specification value whichever is smaller from the local reading at the HMI.
- e) All SCADA setting / Alarm tags shall be user configurable with high level password.
- f) System vendor shall provide the final system settings with justification.

- g) All timers and parameters shall be user configurable with different level password and shall not be hard coded.
- h) System vendor shall provide system alarm list for review and approval before implementation.
- i) Initial system acceptance test (or FAT) shall be carried out at contractor's premises in order to demonstrate operational and functional testing of the system. Though it is carried out at the factory, detailed testing of the I/O list, I/O Loop test, assigned tagging both on component and on HMI, I/O firing test, instrument scaling test, etc. shall be performed. Detailed FAT requirement is explained under Testing and Commissioning section.
- j) All qualification instruments used online shall be with required calibration certificates. Soft copy or hard copy of the instrument measurement values shall be used as the proof for the qualification.
- k) All on-line qualification instruments shall be selected such that the RE% shall be less than 1% (or lower based on requirements) of the intended specification. This shall be validated during design review.
- l) Initial qualification of the Systems shall be performed with Ultrapure Water wherever applicable. It is important to meet the criteria for qualification with UPW at points-of-use associated to chemical blender system. Demonstrate system performance for all Chemical Blending systems using Ultrapure Water (unless otherwise specified).
- m) After successful performance testing with Ultrapure Water and the acceptance from Micron or EPCM company, contractor is responsible for sampling and final qualification of Chemical blending systems with actual chemical. Samples to be analyzed at an approved Laboratory whose analytical services are acceptable to Micron. Other Laboratories can be proposed and quoted as an alternative in the contractors offer.
- n) Contractor to demonstrate system performance of blending, mixing and dilution systems with actual chemical wherever is applicable.
- o) Deliver blended chemical to point-of-use with no degradation outside of the acceptance criteria specification of blended chemical quality. Quality is determined by collecting samples of each blended chemical at the blending system and at POC, and testing the collected samples for assay and metals (and Particles

if requested). Systems are not accepted until acceptance criteria are met.

- p) Sampling by Micron (or appointed TGCM Contractor) will be conducted weekly/fortnightly/monthly during warranty period following initial acceptance of System in conformance with the same acceptance criteria specification to evaluate the consistency of the blended chemical supplied to the Fab.
- q) It is very essential to note that prior to chemical filling, contractor has to provide and comply with Micron safety requirements. This shall include but not be limited to:
- r) Only chemical handling trained workers are allowed to work with chemicals. The employees handling chemical shall be HAZMAT trained.
- s) Safe operating procedure - e.g. spill chemical handling and cleaning procedure, chemical receiving procedure, safety standing order, safety training and emergency response team and procedure
- t) PPE – hard hat, safety shoes, safety glass, arm guard, aprons, acid/solvent gloves, and first aid kit
- u) Spill control facilities - spill absorbent pads, spill absorbent socks, spill absorbent pillows, acid/base/HF neutralizer, pH paper, barricade cones and barricade tape.
- v) Safety signs - correct signage(s) and notice(s) on door(s), proper chemical labelling on tanks and pipes
- w) Fire protection - fire extinguisher
- x) As for solvent, all stipulated code and other ATEX requirements shall be met.
- y) Ensure all the required commissioning activities such as electrical testing, mechanical installation inspections and other integrity checking, SAT, flushing with UPW, etc. are completed before chemical charging.
- z) Seismic Performance: depending on the location of the Fab required importance on compliance to local and international regulations shall be conformed to by the contractor. Contractor shall provide equipment anchoring in conformance to the requirements of the site specification and the applicable local jurisdiction
- aa) A checklist for first charge of blended chemical into system & loop and a checklist for blended Chemical Receiving is attached in Appendix 3 & 4.

6.1.3 General Civil, Structural and Other Utility Requirements:

Though chemical blending system builder is not involved directly in building the chemical rooms, docks, move-in paths, entrance size and other civil /structural works, chemical blending system contractor has certain responsibilities and upon not fulfilling the required responsibilities, contractor shall face the consequences.

- a) The Contractor shall be responsible for checking and monitoring the building with regard to conformance to contractor's civil brief and specifications. If contractor fails in notifying Micron about deviations found in time, he will be responsible for absorbing any resulting costs and schedule delays, except costs for the necessary correction. All construction activities shall be coordinated within the Chemical rooms once the rooms are handed over from Micron to the contractor.
- b) The Contractor shall check the structural drawings before ordering equipment. The Contractor shall prepare plinth and penetration drawings and room layout drawings three weeks after award of the contract. As part of the tender, a wall penetration drawing may be provided for common wall penetrations for the contractors to utilize. However, contractor shall confirm/provide the required wall penetration locations and sizes.
- c) Sump pit location & size and trench sizes for required drainpipe size & slope are critical to avoid flooding, drainpipe choking and backflow to systems. Contractor shall study civil drawings with respect to drain (especially drain quantity during each blended chemical qualification) and collection and notify the Micron if changes are needed. However, as the changes in sump pit size and trench sizes will affect the construction plan, consensus to be reached between Micron and the contractor.
- d) Detailed size and location information of the of the chemical blender systems and related equipment shall be confirmed during the contractor's detail design stage with input from Micron.

Note: If it is an expansion project, civil/structural work may also have to be covered by the contractor based on requirements.

6.1.4 Temporary Utilities and Provisions

- a) Based on the contractual agreement Contractor, Micron (or appointed EPCM) shall provide all necessary containers for pre-fabrication.
- b) Contractor shall supply and install temporary floor covering in each room to protect floor and walls from any damage
- c) Temporary power will be provided by Micron or EPCM for low amperage requirements, i.e. grinders, drills, etc. This temporary power is not guaranteed to be without fluctuations or interruptions and is not suitable for orbital welding. Contractor shall take necessary precautions by installing provisions to keep the power supply stable. The Location of temporary power panel to be advised once temporary power supply is established.
- d) Micron or EPCM will provide Low level lighting for Safety purposes throughout the building. Contractor to provide task lighting in each of the Chemical rooms and/or other required areas. Code compliance for task lighting in all required areas is a necessity as needed basis.
- e) Contractor shall provide mini clean room (or flow box with proper laminar flow) with utilities for pre-fabrication of ultra-high purity piping assemblies, etc.
- f) Based on the requirement for site office and quantity of material to be handled, contractor shall provide on-site 20-40 ft containers for material storage and site offices. Provide utilities, telephones, internet, etc. as required.
- g) Contractor shall provide high quality welding Argon per SEMI F78 for SS UHP piping system installation.
- h) Contractor shall provide high quality Nitrogen 5.0 for drying and pressure test
- i) Due to delay in any utility system installation, required utilities may not be available and temporary utility may have to be arranged for testing and commissioning activities. It is the obligation of the contractor to ensure to bring up the relevant issues with construction team and get it rectified with Micron site team (or appointed EPCM contractor).

6.1.5 House Keeping

Construction management shall form part of the scope of the Contractor. The Contractor shall be responsible for maintaining good housekeeping in all areas where the Contractor personnel works, e.g. storage and prefabrication areas.

- a) Cleaning team shall always be in place to maintain the Chemical rooms and other work areas in a clean and orderly manner. Cleanliness will be closely monitored by Micron for compliance.
- b) If anyone need to enter cleanroom, shall comply with Micron Clean-room protocol specification.
- c) Based on the area of work and based on protocol requirements, contractor shall provide the shoe covers, clean room shoes, facemask, gloves and cleanroom garments to the contractor employees.

6.1.6 Electrostatic Discharge:

Buildup of static charge in the liquid medium is a common phenomenon in fluid movements due to the shear between surfaces and liquid. When the liquid is conductive and has a path for neutralizing the charge, static charge buildup can be eradicated or reduced. When the flash point of medium (liquid) is lower, the impact of the static buildup can be catastrophic. The following points shall be critically given importance to avoid issues.

- a) As for low flash point solvents, ensure charge neutralizing methodology is implemented.
- b) If pressure at any point of the system goes to vacuum (below atmospheric), a detailed study shall be performed for vapour pressure and its relevance to any catastrophic incidents.
- c) All metallic components of a solvent distribution system shall be bonded together to form a common low impedance path to the electrical grounding system.
- d) Explosion-proof components shall be designed and utilized for organic solvent systems, according to safety and code requirements for the specific chemical(s) utilized during the design phase.

6.1.7 Failure Mode and Effect Analysis and HAZOP

Failure Mode and Effect Analysis (FMEA) will have to be carried out by the system supplier in separate FMEAs for Process step functions. If a separate FMEA to be done for controls for having better understanding of different failure modes due to controls, it shall be done. Process FMEA shall be carried out in a structured approach that all potential failures that may exist in the system to be clearly discovered.

As minimum requirement, Process FMEA shall cover the following:

- a) All different processes such as rinsing, filling, stand-by, ready, etc.
- b) Target set points, Specifications, control limits.
- c) Process qualifications based on different online parameters
- d) Process parameters such as tank levels, loop pressures, differential pressure, conductivity, concentration, density or Specific Gravity, temperature, etc.
- e) Deviation limits between two instruments when two instruments are used to do monitoring of each other
- f) Safety parameters such as leakages, rated pressure, exhaust, air exchange rates, etc.
- g) Other safety parameters such as UV/IR detection, etc.
- h) Bubbles in suction line
- i) NPSH and vapor pressure of the liquids
- j) Degassing nature of the liquids
- k) Vapor pressure of the liquids and the required N2 blanketing pressure
- l) Utility requirements such as UPW flow and pressure, Instrument air & CDA flow and pressure, UN2 flow and pressure, Process N2 flow and Pressure, etc.
- m) Based on FMEA (Link: [Facilities Process FMEA and Control Plan Business Rules](#)) outcomes, if there are any routine activities to be carried out such as preventive maintenance of equipment, calibration/validation of instruments, health check of any components, etc., these activities have to be specifically stated with required procedures.
- n) Before any installation of any system, Relative Risk Assessment (Link: [RRA Procedure](#)) according to Micron's Process Safety Management shall be carried out and

when the Relative Risk Rating falls in dangerous or Hazardous categories according to Micron's Process Safety Management, HAZOP (Link: [HAZOP Study Business Process](#)) shall be carried out. When HAZOP is carried out, it is the duty of the system supplier to arrange facilitators, experts of different disciplines, Micron representatives, etc. to drive the whole required processes of HAZOP.

6.2 Design Concept and Requirements:

6.2.1 General Design Requirements:

- a) Chemical room sizes and layouts shall be coordinated with Chemical dispense and blender system designs to maximize the number of systems installed in any one room. Space allocation for future dispense or blend systems shall be evaluated during the design phase.
- b) All parts of the systems made of plastic materials working under pressure shall be protected by double containment. Double containment shall be able to contain the chemical with proper sealing until a proper response is taken.
- c) Based on risk assessment, code analysis, and safety requirements, sprinkler systems shall be installed in all chemical system rooms.
- d) Chemical dispense and blender modules in closed-cabinet configuration shall be equipped with exhaust according to the chemical properties and safety requirements. Materials for the exhaust ducts shall be identified based on compatibility, risk assessment, and cost comparison. These materials are typically Teflon-lined S.S. duct work or other FM 4922-compliant materials. All N2 blanketed vessels shall be connected to the exhaust, and related vents shall be routed to the appropriate exhaust system.
- e) All chemical dispense and blender systems shall be in dedicated rooms rated for the type of hazard identified with the respective chemical.
- f) Material routing and equipment move-in path shall be designated during layout planning.
- g) Day tanks shall be routed to the appropriate exhaust system with some type of check valve system to prevent back contamination from the exhaust system.
- h) Space for future expansion and additional chemical dispense and blender systems shall be planned during the initial room design and therefore, system footprint and space between systems shall be optimized to maintain

space for future systems. Also, ease of removal of tanks and systems shall be taken into consideration where possible during room layout planning to minimize land locking.

6.2.2 System Design Parameters:

- a) Generally required Chemicals are supplied either from transfer unit or from distribution network.
- b) All blended chemicals shall be supplied to distribution system or distributed to Fab. When blended chemical is distributed to Fab, requirements stated in [Global Facilities - Design & Construction Standard - Chemical - Chemical Distribution System](#) shall be strictly followed.
- c) During the entire transfer path, the blended chemicals must be protected from contamination and/or change from the blend properties.
- d) Review of chemicals and corresponding chemical blender systems shall be completed, and appropriate classifications assigned to each such as corrosives, oxidizers, water reactive, toxic, etc. for inorganic chemicals. Organic chemical classifications require review of chemical properties such as flash point and boiling point to determine IFC (or equivalent adopted code) Hazard Classification. Class and division require further review to determine system design and control parameters. This review process shall be completed early in the design phase and shall be used to determine applicable code requirements and compliance, system design features, system materials of construction, system placement within rooms, and room classifications.
- e) Based on requirement, chemical blender systems shall be designed to operate continuously, with the theoretical capability to operate 100% of the time. Redundant components shall be provided to guarantee the supply of chemicals even if part of the system shuts down due to a failure or maintenance.
- f) Based on requirements, the delivery pressure shall comply with the limits and parameters provided by Micron.
- g) The delivery pressure shall be evaluated at the farthest Point of Connection (POC) and the end of the distribution line in a dead leg system.
- h) Chemical blender system piping shall be sized based on system flow requirements and design parameters.

- i) If N2 is used for blanketing, ensure safety, the N2 blanketing design must avoid the Explosion or Implosion of any tank or other components.
 - i. To prevent explosion, the N2 blanketing system shall not exceed the maximum pressure of any tank or any other component at all times. Safety relief valve or burst disk shall be sized and installed according to the type of drum, day tanks and bulk tanks and in accordance with all applicable codes and industry standards.
 - ii. To prevent implosion during a transfer (drum to day tank or blend tank to day tank) or dispense (chemical consumed by tools) mode, blanketing N2 flow shall compensate the maximum liquid flow.
- j) To prevent contamination, the blanketing system must guarantee that the N2 flow compensate the liquid flow during a transfer or dispense operation.
 - i. The tanks shall be prevented from back contamination from exhaust (due to insufficient N2 flow), by designing the system based on the maximum liquid flow transferred.
 - ii. Based on requirements, tanks shall be routed to the appropriate exhaust system with a check valve system and/or filter to prevent back contamination from the exhaust system.
- k) If relief valves are utilized for N2 blanketing tanks or any other components, it shall be sized in accordance with all applicable codes and industry standards.
- l) Chemical blending modules shall be constructed as enclosed system in which the pumps and pressurized components are contained within a chemical compatible enclosure based upon local codes and regulations, safety requirements, principles of functionality, chemical properties, available space, and design criteria.
- m) Non-wetted materials shall be identified according to safety requirements, material compatibility, risk assessment analysis, and cost comparison.
- n) The maximum rated pressure of all components shall be considered during the initial design. Design shall consider all local codes and regulations, safety requirements, and operating conditions.

- o) When systems are required to operate with a specific temperature range, appropriate temperature control and monitoring devices shall be installed.
- p) Each module shall have the appropriate facility utilities made available within 3m of the system. Typical required facilities for a blender are: Power (Normal Power and UPS (Uninterrupted Power Supply), UPW, Chemical(s), Instrument Air, CDA, UN2, NPW, drains, and exhaust. Refer to Appendix 5 – Utilities Interfaces and Requirements.
- q) The following parameters shall be evaluated and combined to model the most efficient chemical blender system with defined capacity:
 - i. Recipe of the blend. Specification limit and control limit shall be set, and control limit shall be used for system qualification.
 - ii. System Design Considerations:
 - Total quantity and type of blenders to be supplied.
 - The length of the distribution line(s) from Blender to chemical transfer unit, CDM and Day Tank of CDM.
 - Chemical room space and positioning.
 - iii. Tools flow requirements:
 - Average flow: is based on Tool Service Requirements (TSR) data for each tool. Average flow is the amount of blended chemical required by the tool to flow from the POC to the Point of Use (POU). The sum of the average flows of all tools is used in the calculation to determine the day tank volume.
 - Peak flow: is based on Tool Service Requirements (TSR) data for each tool. Peak flow is the amount of blended chemical required by the tool to flow from the POC to the Point of Use (POU). The sum of the peak flows of all tools is used in the calculation to find the system diversity factor and ultimately the system maximum design capacity, distribution loop size and design flow.

- iv. Diversity factor: is a derived number typically between 0.01 and 0.99 based on the tool utilization rate of blended chemical compared to actual consumption. To find the diversity factor for a particular system, the following items need to be taken into consideration: peak flow of all related tools connected to a chemical blender system and tool cycle time, wafer throughputs, tool utilization percentage, and other factors.
- v. Design flow to be guaranteed at the required pressure. It is calculated by multiplying the peak flow by the diversity factor and shall guarantee to correctly supply POC at the required pressure and flow.
- r) System modeling, using PIPE-FLO®, AFT Fathom 9.0 hydraulic or an equivalent software program, shall be used during distribution network design phase to review system dispense and distribution loads, sizing, and minimum performance requirements. CFD or an equivalent software program can be used for other modelling such as mixing, tank Low-Low level analysis, pump suction head analysis, and other appropriate design analysis.
 - i. When a new CDM is added to the distribution system, the system piping model shall be re-engaged to ensure that the impacted blender and subsequent distribution system can support the new addition. The model shall also be utilized to understand any system impacts due to the changes, and how to mitigate those impacts.
- s) When modifications or additions are required to be made to an existing operational system, those modifications shall be tested and qualified prior to installation and shall not be physically connected until they are ready for operation.
- t) Construction Material shall be followed as below: When contractor realizes the material, specification is not appropriate for the specific chemical, it is the duty of the contractor to inform Micron in writing. Thus, contractor also takes ownership of material selection.
- u) To prevent excursion and minimize response time for operation validation, refer to Global Facilities - O&M BKM - Chemical - System interlock Design and Improvement Guideline that shall be incorporated into system interlock designs.

6.2.3 Blender System Design Configurations

- a) Blended chemicals are obtained by mixing one or more UHP concentrated chemicals with or without UPW. The dilution process shall be carried out avoiding any contamination of the concentrated chemical(s), the UPW, and the final diluted product. The diluted chemicals shall be controlled to guarantee the exact concentration within the control limits and parameters identified by each site's Facilities Chemical Engineer or other chemical system owners.
- b) Blending systems shall be capable to automatically blend multiple chemical streams which are continuously monitored by control systems for proper flow, pressure, or quality. The control system shall be programmed to automatically adjust the chemical streams in the event of a flow, pressure or chemical specification deviation. If system cannot control the blend stream it will divert delivery to the appropriate waste drain or stop the blend and alert the Chemical Operator.
- c) Chemicals shall be blended based upon, but not limited to, the following methods and/or a combination thereof. The best method shall be identified in the design phase, based on interface to the chemical dispense system, type of blended chemicals, concentration limits, metrology accuracy, reliability, and performance.
 - i. By conductivity.
 - ii. By flow and conductivity.
 - iii. By weight
 - iv. Or other specialty metrology/analytical devices approved by site Engineering.
- d) Recommended blending methodologies are In-line, batch blending and combination of both in-line and batch. A batch process offers the ability to integrate metrology to validate concentration before transferring to a day tank/storage tank but with limited capacity, but an Inline process can offer higher capacity and, possibly, the ability to vary the concentration based on the needs of the process. Some considerations with respect to the methodologies are given below:

1. **In-line Blending Method:** Blending concentrated chemicals with or without UPW (2 or 3-part blending or more) through flow controls with qualitative specification review prior to distribution to final product tank. There shall be a redundant stream to mitigate against single point of failure. Inline blender shall have both Feedback (closed loop) & Feed-Forward (Open Loop) blending user configurable options. Refer to **Appendix 8 & 9** for Typical P&IDs of an Inline Blender.
 - i. Feedback (closed loop) Controlled Blending: Blending process which uses blended chemical information, such as assay or conductivity, as the basis for controlling blending of constituents through some type of automated flow controller device. This method is utilized where processes that require metrology devices to validate the concentration of the blended chemical and/or to provide a feedback blending capability. Feedback blending utilizes the concentration data from a metrology device for a given blend to dose to achieve the target concentration.
 - ii. Feed-Forward (Open Loop) Blending: This blending method is based on the metrology set and will commence immediately upon demand and send product directly to day tank or storage vessel. This method assumes that the initial flow passing through the system metrology may or may not be in specification, however based on the low volume in comparison to receiving vessel volume, out of specification material will not impact the overall performance of the deliverable product. This may be an acceptable practice based on criticality of product, and reliability of blender. This practice allows for some cost savings by forgoing the typical blend practice of running blended material to drain for X period time, where X is a settable parameter and is based on metrology performance and response time.

2. **Batch Blending Method:** Blending concentrated chemicals with or without UPW (2 or 3-part blending or more) into a batch tank with precise load cells to measure the weights of constituents in a batch process. The blended chemicals are recirculated in a closed loop and monitored via some proven metrology (such as conductivity). Blend remains within a batch tank until blend falls within specified control parameters. If blend does not reach control parameters, system will auto adjust if configured as an option (based on batch specifications & program parameters) until it reaches reference value. Once blend parameters are reached, the approved batch is then transferred to the blended chemical dispense system day tank or storage tank. To mitigate against single point of failure, batch blenders shall have a minimum of three batch tanks. Refer to Appendix 10 for a Typical P&ID of a Batch Blending system.
 3. **Combination of both Batch & Inline Method:** A blend system which utilizes both Inline & Batch blender components and configurations might be needed based on chemical properties, high dilution ratio and high accuracy of one or more constituent(s) of the final blend target concentration. The inline method if utilized can be either using flow measuring devices that totalize flow or precise flow controller to control volumes of constituents or using a metering pump (pump strokes counted and volume calculated based on volume per stroke) which is finally added to a batch tank with load cell.
 4. **Special Combination Method for 3-part Blending:** For CMP chemicals like KP3100 & STI2910 where forming is a concern, two separate blending methodologies shall be utilized to achieve the final blend target. A standalone batch blender shall be utilized for KP3100+STI2910(1:1) mixing for the first of 2-part blending and once qualified will then feed a standalone Inline blender where it blends with UPW to achieve the final target concentration value. Refer to Appendix 11 for a Typical P&ID of KP3100+STI2910 Batch Blending System.
- e) When blending solvents that are classified as flammables and/or combustibles all applicable code requirements shall be reviewed and met during the design phase.

- f) The following parameters shall be considered and combined in the design phase:
 - 1. Flow requirements:
 - i. Daily average usage (LPD or GPD).
 - ii. Peak flow: the maximum flow required from all tools.
 - iii. Peak flow duration time; the duration time of the peak flow condition.
 - 2. Volume capacities of the day tank(s) and/or storage tank(s).
 - 3. The minimum sizing of distribution line from blender to day tank/storage tanks shall be 1" tubing. Final sizing shall be derived during the design phase based on the flow requirements.
- g) The blended chemicals shall be transferred to the day/storage tank(s) at intervals that will maintain the highest level within the tanks in order to avoid interruption in the case of failure of the blending system.
- h) When filters are installed, the maximum Delta P through the filter assembly shall be recorded and alarm parameters set accordingly. Proper values shall be based on chemical properties, consumption, tool process and design criteria, and monitored via local and SCADA systems.
- i) Blending process parameters fluctuation shall be controlled to specified limits (or RE $\leq 1\%$ of the specified limits if RE % is not specified) or as required by blended material target specification and control limits.
- j) Chemicals used in the blending process shall be supplied by gravity, pumps, UHP N2 pressurized system, or by vacuum generation, based on dilution ratio, accuracy, design criteria, and safety requirements.
- k) The chemical blend systems shall be equipped, but not limited to, the following components and features:
 - 1. Chemical blending tank/vessel (depending on blend design selected).
 - 2. Isolation valves (manual and pneumatic), Needle valves and flow control valves, that meet UHP chemical standards, and are compatible with chemicals in-use.

3. Centrifugal, Metering and Pneumatic pumps. In the case of pneumatic pumps, a pulsation dampener shall be equipped downstream of the pump outlet to mitigate against pressure fluctuation and avoid filter damage. For more details of centrifugal & pneumatic pump requirements, refer [Global Facilities - Design & Construction Standard - Chemical - Chemical Distribution System](#).
4. Batch or Inline Blender - Dual conductivity probes (or any other dual metrology) used for qualification shall have deviation monitoring feature with user configurable setpoint and warning alarm in the event of drift or failure of one conductivity probe.
5. Batch blender shall have continuous or interval based (user configurable) local recirculation feature for qualified/standby batch to monitor the quality of the batch through a metrology (conductivity or concentration meter, etc.). The alarm setpoints shall be user configurable with deviation warning alarm. There shall be a redundant set of dual metrology for concurrent monitoring of both standby blend tank as well as for blend tank that is being qualified.
6. Batch Blender – Transfer and Blend time-out warning alarm with user configurable setpoint.
7. Batch or Inline Blender – Number of auto-correction during blending shall be user configurable with a warning alarm and when the set auto correction is exceeded, it shall prompt operator to either select “resume auto correction” or “abort”.
8. Batch Blenders shall have the feature to automatically drain the tank and will prompt operator to allow tank draining and resume auto correction or abort.
9. Critical Actions - “Pause”, “Manual” “Stop”, “Abort batch”, “auto correction”, “Offline” and “Disable” shall have pop-up confirmation button at HMI for re-confirmation of action and shall have pre-warning alarm at HMI/SCADA.
10. Normal Actions - “Resume”, “Auto” and “Online” shall have pop-up confirmation button at HMI for re-confirmation of action and shall have Event log at HMI/SCADA.
11. Conductivity and Assay(wt%) values of qualified batch shall be displayed in HMI.

12. Trending for all critical metrology parameters used for qualification like Assay, conductivity and metrology probe temperatures shall be trended in HMI & SCADA.
 13. Pressure transducers
 14. Flow controllers.
 15. Flow meters.
 16. Filter manifold(s) (if required).
 17. Proximity sensors for level/refilling control, N2 bubbler, external level probes, radar, ultrasonic or other reliable level control devices.
 18. Temperature transducer when temperature control is identified in the design phase.
 19. Static mixers or other devices that promote a homogenous blend.
 20. Metrology equipment – conductivity, concentration, etc.
 21. Scales, load cells etc.
 22. Pressure/Back pressure regulators.
 23. Eductors
 24. Flow dampeners
 25. Heat Exchangers
- l) The chemical blender system shall be equipped with spool pieces made of welded parts and coupled fittings to facilitate safe and easy maintenance.
 - m) Key system and blending components shall be arranged in a fashion that facilitates removal and replacement in 2hrs or less. Examples of key components are flow controllers, metrology devices, load cells/strain gauges, filters, etc.
 - n) Blenders shall be designed with a minimum capability to blend 2 constituents with one of the blending lines capable to blend at low flow/volumes based on the blend ratios.
 - o) Blenders shall have appropriately sized and chemical compatible pressure regulators and/or flowmeters installed prior to flow metering/control device. In some cases, back pressure regulators may be required after the flow metering/control device depending on the design requirements.

- p) At minimum blenders and key blender components shall be capable to provide blended material at 1.5X greater than average daily consumption.
- q) Blenders shall be configured such that they are capable to blend and transfer to a minimum of three locations:
 - i. bulk storage tanks
 - ii. day tanks
 - iii. other vessels such as totes or drums
- r) In-line blenders that require concentrated chemical or UPW pressure to transfer blended material to storage vessels the chemical/UPW pressure shall have sufficient flow & pressure to transfer the material at the required rate to meet the maximum consumption rate. Depending on high flow/capacity requirements of the Inline blender, the UPW supply pipe shall be sized appropriately to meet the demand needs and in the event of multiple high flow Inline Blender systems are used, separate UPW lateral shall be used for each system.
- s) Systems designed to batch blend shall have load cells/strain gauges/scales that are capable of accuracy ranges of ± 0.01 lbs or 0.005kgs or better depending on the accuracy/spec requirements of the final blend.
- t) Batch blender designs shall incorporate the appropriate in-line, on-line, in-situ metrology devices so as to provide qualitative data that indicates the batch material is within specifications and allows for the shortest batch blending times prior to release to blended chemical distribution system.
- u) When blending more than one type of chemical blend through a single blender, all blend lines used for Blend 1 must be rinsed with UPW or in the case of solvents the main constituents signature spec prior to start of Blend 2. Typically, this is achieved by running UPW or proper solvent to drain until it reaches rinse material designated specification.
- v) In-line blenders shall be designed so that upon completion of the blending run the system will automatically perform a clean-up routine whereas the system rinses the appropriate plumbing and metrology components back to designated starting specification. To determine achievement of starting specification an appropriate on-board metrology device shall be utilized.

- w) Blender design shall take into consideration any redundancy needs that would be critical for continuous blender operation. The following items should be (but not limited to) included during design consideration for redundancy needs:
 - i. key blending control device(s) i.e. load cells, conductivity probes, flow controllers, etc.
 - ii. Three batch blending tanks.
 - iii. recirculating pumps on batch blenders (if utilized).
 - iv. if filters are installed within blending system N+1 housings shall be added based on max flow requirements, ΔP impacts and requirements for periodic filter maintenance.
 - v. level sensors and probes.
 - vi. A secondary redundant blend line with static mixer for in-line blenders.
- x) When consumption drives the design to have two or more blenders for the same chemical blend, blenders shall be cross connected so that each can fulfill the other's responsibilities.
- y) Where two or more blenders are in place, and blenders are functionally similar, and chemical blends shall be chemically compatible, blenders shall be designed so that each blender has the ability to provide back-up blending duties to the other similar blender.
- z) Blender expansion shall be given due consideration:
 - i. Where flow controllers are employed, the ability to change the flow controller size.
 - ii. A main branch valve shall be supplied on the bottom of the blending tank, and when utilized for expansion, another valve shall be equipped to drain and/or expand the supply of blended chemical (Take one, leave one philosophy).
 - iii. Filter manifold shall be capable for expansion (if installed).
 - iv. Cabinet/enclosure shall be designed to allow for expansion of a second blend leg or other metrology equipment.
 - v. Batch systems controls & plumbing shall be designed to expand to allow for feed and/or monitor two or more day tank/storage tanks.
 - vi. Heat Exchanger manifold (if installed) shall be capable for expansion.

- aa) A summary of cabinet, piping and internal component material by media category for chemical blender systems are shown in Table 1 below.
- bb) For In-line blenders, Flow Rates shall be user configurable based on flow controller working ranges and blended chemical demand. Similarly, for batch blenders, batch size shall be user configurable based on the demand.
- cc) Waste generated during blending (or initial stabilization) and during transfer needs to be reduced. In-line blenders need to be stabilized within one minute and flowrate needs to be optimized so that the system start/stop frequency is low.

Table 1 - Summary of Cabinet, Piping & Internal Component Material by Media Category

Media Category	Cabinet Material	Piping/Internal Components in Cabinet
Acid / Caustic	PP/PVC with clear PVC viewing panel	- Piping: PFA 450/451HP or AP231SH PFA - Components: Wetted parts shall be PTFE /PFA / PTFM
HCL / HF	PP/PVC with clear PVC viewing panel	- Piping: PFA 950/951HP or LPP PFA - Components: Wetted parts shall be PTFE /PFA / PTFM
Flammable Solvents	Polished stainless steel with anti-static viewing glass panel.	- Piping: Static Dissipative PFA or SS 316L-EP (RA=8). - Components: Wetted parts shall be SS 316L-EP (RA=8). SS valves shall be with weep hole (leak port).
Flammable corrosive Solvents	Polished stainless steel with anti-static viewing glass panel.	- Piping: Static Dissipative PFA or Equivalent - Components: Wetted parts shall be PTFE /PFA / PTFM
Non-Flammable Corrosive Solvent	Polished stainless steel with viewing glass panel.	- Piping: PFA 450/451HP or AP231SH PFA - Components: Wetted parts shall be PTFE /PFA / PTFM

6.2.4 Day Tank Design and Configuration

Refer to [Global Facilities - Design & Construction Standard - Chemical - Chemical Distribution System](#) for additional requirements with respect to design, design configurations, nozzles, spare ports, isolation valves, safety devices, instrumentation, etc. that are related to tanks.

6.2.5 Chemical Room Sump System

Refer to [Global Facilities - Design & Construction Standard - Chemical - Chemical Distribution System](#) for additional requirements with respect to design and configurations.

6.2.6 Power Distribution Design Criteria:

The Design of Power Distribution System for chemical blender systems is very important for continuous operation of the chemical blender systems and therefore, shall be designed to provide a continuous power supply to chemical blender and blended chemical distribution equipment and the respective controls systems. Power Distribution System shall mainly consist of Main Power Distribution Panels, Room power panels and all required cabling and connections to each piece of equipment. The chemical blender system contractor shall supply the Main Power Distribution Panels for termination of the feeder cables based on requirements. The room power panels can be placed generally outside the chemical rooms unless any other location is specified due to space constraints.

The AC Main Power Distribution Boards are the main receivers of incoming power for most of the Electrical and Control systems. Incoming normal utility power shall be continuously monitored in the Panel. Any anomaly such as voltage sags will cause the internal circuitry to command to transfer to the standby source while additional switch circuitry determines the source has the proper voltage and frequency. When normal utility power returns or no anomalies have occurred, the operator can then manually transfer back to utility power when it is deemed appropriate. When one bus is under maintenance, other bus which supplies power through dual power supply maintains power in the panel and thus to chemical blender and blended chemical distribution systems.

Following points shall be generally considered in the power distribution design for the chemical blender systems.

- a) Power distribution design to address electrical redundancy and single-point-of-failure mitigation.
- b) Any AC CDU or AC stirrer directly supplied from the Dual Feed AC Main Power Board requires dual feed power supplies.

- c) Points of connection (By Others) for Normal Power and UPS/CPS (Uninterrupted/Continuous Power System) power will be provided in the area designated for the process Chemical (Blender and Distribution) Systems and terminated at fusible disconnect switches in the Electrical Room – chemical (Blender and Distribution) system contractor shall liaise with electrical contractor to ensure the feeder cables are terminated in the chemical (Blender and Distribution) electrical panels as per agreed design.
- d) Connect critical equipment (i.e. cannot withstand prolonged power interruption) to UPS/CPS power. There will be no interruption to UPS/CPS in the event of a momentary or prolonged Normal Power outage. However, the contractor shall have to make sure how critical equipment will be in operation in the event of power supply lag. Equipment connected to Normal Power will experience a power loss in a Normal Power outage and remain offline until Normal Power is restored.
- e) As power source supply from grid is 400 VAC, 3 phase, 50 or 60 Hz (depending on region/country), provide transformers and panel boards to derive 380V, 200-240V/100-120V, 24VDC or other 3-phase and single-phase sources.
- f) Furnish and install motor control centers, VFD's, instrument power transformers, electrical enclosures, equipment power disconnects, conduit, and cabling for proper operation of the System
- g) If Solvents are used, electrical for solvent collection and transfer operations shall be Code compliant.
- h) Provide protective devices within the Chemical (Blender and Distribution) Electrical equipment to prevent the propagation of electrical faults generated within the Chemical dispensing rooms or within Chemical (Blender and Distribution) dispensing or distribution equipment
- i) Label electrical equipment (panels, transformers, motor control centers), conduit and wire per the Equipment Coding specifications in the Electrical Specification Design Standard.
- j) Contractor to follow project Ferrule requirements as specified in the Electrical Specification Design Standard.
- k) Contractor to follow project electrical panel (MCC's) spare requirements as specified in the Electrical Specification Design Standard.

- l) Future Circuits: 20% spare or as specified in the single line diagram, whichever is greater.
- m) The Electrical distribution will be modeled in a Power Systems Study Program by others. Prior to equipment installation. Contractor to provide loading data for panels and equipment within the Chemical dispensing rooms for incorporation into the database. Power Systems administrator will review submitted load data to ensure panels are not overloaded.
- n) Do not install liquid piping immediately over or within a 3-foot plan view clearance of any electrical panel, motor starter, or control panel. Where piping must be located within these zones, either install piping inside a PVC conduit or shield the electrical device to prevent direct liquid access to electrical equipment. Locate piping such that bottom of pipe (or pipe covering/shield, if pipe is covered/shielded) is not less than a clear height of 6 feet, above the highest part of electrical equipment located directly below the pipe.
- o) Ingress Protection requirements for Main Power Distribution Panels, Room power panels, electrical panels for individual chemical (Blender and Distribution) systems or other related equipment, motors, etc. shall be according to the zonal specifications stipulated in Electrical, Control, and Monitoring Specifications.
- p) Following electrical design configurations and parameters shall be observed in the electrical design with respect to chemical (Blender and Distribution) systems:
 - i. Chemical systems that are connected to direct dispense tools and all systems where continuous control is required shall be equipped with centrifugal pumps or high volume dual pressurized vessels. The pumps and vessel controls shall be powered from two separate power sources to provide maintainability (A/B philosophy): a normal power source and a UPS source.
 - ii. All other chemical dispense systems used in tool bath filling within a cycle time and/or intermediate reservoir tank filling dispense systems can be equipped with centrifugal pumps, pneumatic pumps, or pressurized vessels. The pumps and vessel controls shall be powered from two separate power sources to provide maintainability (A/B philosophy): a UPS source to one pump and a

normal power source to the other; or, a source backed by sag mitigation to one pump and a normal power source to the other.

Note: Power Distribution Panels (PDPs – 24VDC) or Power Distribution Units (PDUs – 24VDC) are not recommended for power to chemical (Blender and Distribution) systems. If they are utilized, they shall have redundant power supplies that can automatically switch over upon a failure and the incoming power supply shall have SAG mitigation or a UPS feed.

- iii. For protection from power disturbances, the chemical dispense systems controller utilized for bath filling within a cycle time and/or intermediate reservoir tank filling dispense systems shall be powered using either a UPS source or a source backed with a SAG mitigation system.
 - iv. All metallic components of a solvent distribution system shall be bonded together to form a common low impedance path to the electrical grounding system.
 - v. Explosion-proof components shall be designed and utilized for organic solvent systems, according to safety and code requirements for the specific chemical(s) utilized during the design phase.
 - vi. Power requirements such as UPS, SAG mitigation, and normal power for chemical systems and components shall be applied based on system and tool needs.
- q) Refer to Electrical Specification Design Standard for additional required scope of work. Link: [Global Facilities-Electrical Specification Design Standard](#)
- r) Monitoring and rectification shall be made possible when any power supply component fails.
- s) If there is any discrepancy between the above electrical requirements and specifications mentioned in Electrical Specification Design Standard, specifications shall be followed after obtaining approval from Micron.

6.2.7 Instrumentation, Control and Monitoring

The Chemical (Blender and Distribution) Instrumentation, Control and Monitoring System shall be designed to control, monitor and report leak sensor activity for the distribution piping networks, chemical (Blender and Distribution) system activity (events) and alarms, room sumps, and as well as process tool requests based on the requirements. The main components shall consist of process instruments, chemical (Blender and Distribution) system PLCs, Data Concentrator(s), a SCADA terminal, Field I/O Panels (RIO), Master and Room EMO panels, sump station controllers, an Ethernet cable network for communication between the various control system components, and a network for communication with various operation stations. Brand & Model of PLCs, acceptable brand for I/O & Other Cards, Power supply unit and OS Platform and version of SCADA shall be advised by Micron. SCADA shall be located in the TGCM Room. The Field I/O Panels shall be located as required by VMB requirements and tool requirements. Following points shall be observed for design and design configuration of instrumentation, control and monitoring:

- a) The Blending systems shall be equipped with a Programmable Logic Controller (PLC) to execute all sequence of operations, and a Human Machine Interface (HMI) to allow interfacing with all set points, alarm thresholds and loop tuning parameters.
- b) Data Concentrators used for control & communication between CTU, CDU, distribution network and tools (if any) shall be used for respective control and communication related to blender systems.
- c) A "System Available/Not available" status signal shall be available to the dispense systems.
- d) A "Blend demand" signal shall be included in the PLC input, as feedback of the chemical blender system based on dispense system usage.
- e) All analog signals and alarms shall be monitored & historized by the site's FMCS or SCADA system.
- f) Based on the blending methods identified, the PLC shall be interfaced to the load cells, conductivity meters, flow controller, or any other field mounted instrument utilized to control the blending process.
- g) All analog devices, and associated hardware & software installed in the blending system that can affect the blend outcome, shall have at minimum the accuracy capability to achieve the blend specifications.

- h) The blending parameters shall be programmable in the PLC through the HMI allowing the owners to control the blending process, PID, limits, thresholds, alarms and other strategic parameters.
- i) HMI shall have the display for assay (concentration), conductivity, temperature and all the other critical parameters. Option to select the target parameter (concentration or conductivity) shall be provided for the operator to select the basis for blending.
- j) All analog signals and alarms shall be monitored and historized by some type Facilities Monitoring Control System (FMCS) or System Control And Data Acquisition (SCADA) system. The following information shall be provided as a minimum on the SCADA screen:
 - i. Individual system status.
 - ii. Network and PLC communication status.
 - iii. All system alarms and EMO.
 - iv. Individual equipment P&ID's.
 - v. VMB/MHEB and other distribution box leak sensor status.
 - vi. Valve Box and Tool activity
 - vii. Monitoring & Control of Chemical blender systems
 - viii. Chemical consumption monitoring and reporting.
 - ix. Preventive maintenance monitoring and reporting, such as valve cycles, filter hours, pump stroke counts, analyzer and instrument calibration intervals.
 - x. Room sump leak detection (status).
 - xi. System performance trending.
- k) All instrumentation and control equipment shall be utilized within the manufacturer's recommended operating parameters.
- l) Dual Leak detection shall be provided within the Chemical blender system and response strategies shall be identified based on site-specific safety requirements, chemical types, and local/national codes and regulations.
- m) PLC and instrumentation cabinets/panels shall have exhaust and Z- purge incorporated into the system design to prevent and control corrosion of electronic components from fuming chemicals such as HCl, HF, Nitric, etc.
- n) Metrology shall be monitored with the SCADA system, and related data returned for trending and troubleshooting.

- o) All controls used in organic chemical systems (solvents) must be rated and/or approved for use in flammable/combustible environments and shall be connected based on local electrical code requirements.
- p) Temperature transducers or thermocouples that are compatible with corrosive or organic chemicals and environments shall be used when temperature control is identified during the design phase.
- q) To prevent filters from clogging, the Maximum Delta Pressure (ΔP) through the filter assembly shall be monitored via the pre and post filter pressure transducers. The PLC will have the ability to calculate ΔP and shall be settable within local PLC and SCADA alarming protocols. The ΔP set point shall be identified based on chemical properties and consumption, tool process, and design criteria.
- r) Pumps are switched over based on settable timers built into the PLC program or other means. For centrifugal pumps, a pump start sequence shall be implemented with a dedicated sequence of operation to prevent abnormal pressure fluctuation.
- s) Signal Interface – CDM/CTU to Blender
 - i. The Chemical Control System requires certain inputs from the Blenders (as Points of Use) to function properly.
 - ii. Each blender communicates with CTU and/or CDM via Chemical Control System (data concentrator) through hard-wired or communication interface.
 - Chemical Demand or request Signal
For each chemistry stream utilized by a blender, a Chemical Demand signal is required; if a blender uses the same chemical from two sources (i.e., 2 CTUs or CDMs), a separate Chemical Demand signal is required for each source.
- t) Control Panel and Enclosure Fabrication Guidelines
 - i. Refer [Global Facilities - Design & Construction Standard - Chemical - Chemical Distribution System](#) for requirements with respect to control Panel and Enclosure Fabrication Guidelines. Refer to I&C Design Standard and Electrical Specification Design Standard for additional required scope of work.
Link: [Global Facilities-Electrical Specification Design Standard](#)
Link: [Facility Management and Control System \(FMCS\)](#)

7.0 Labeling Specification

Safety Signages and Labelling of Chemicals / Chemical (Blender and Distribution) systems and components are very important because they contain information on them about usage, direction, application and storage amongst other equally important information.

7.1 Labels – Equipment / Components

Printed labels shall be used on all required systems and components. Safety warning label shall be provided by equipment suppliers with hazards and the safety precautions identified to ensure safe operation and maintenance of the equipment. These requirements shall fulfill relevant section in EN61010 - Electrical warning labels, relevant sections in Semi S2 and S1 - Hazard Warning Labels & Hazards Labels, and IEC 61310 Safety of Machinery – Indication, Marking and Actuation. Additional labeling guidelines for TMAH >1% shall be adhered as per [Global Facilities TMAH Standard](#).

Required color code shall be used and there shall not be any deviation that can cause any misleading. As a minimum, the following shall be labeled:

- a) Chemical Blending (and Distribution) Systems.
- b) Tanks – While the standard requirements shall be met to ensure the compliance to hazardous material, tanks shall also have the name plates for the tanks.
- c) System Doors and Removable Panels.
- d) Any other cabinets.
- e) Labelling for electrical cabinets and cable trays shall follow electrical requirements/code/standards.

There shall be required tags for the following components:

- a) All Pumps, blenders, mixers, Valves, Regulators, etc.
- b) Pressure gauges, pressure sensor/transmitter, flow sensor/transmitter, flowmeters, etc.
- c) Quality measuring instruments such as conductivity, density, concentration, pH, etc.
- d) Heat exchangers, dampeners, humidifiers, etc.
- e) Tagging for electrical components and cables shall follow electrical requirements.

7.2 Labels - Piping and tubing:

- a) Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.
- b) Pipe Label Contents: Include identification of piping service using the Sub-System Identifier and an arrow indicating flow direction.
- c) Direction of Flow: Where applicable, arrows will indicate the direction of fluid flow within the pipe or system. Arrows in both directions shall be used to indicate systems having bi-directional flow.
- d) Labels shall be placed such that they are visible from whatever angle(s) or positions a person might be expected to be looking at the pipe/system.
- e) Labels shall be placed within close proximity of valves and branch line intersections.
- f) Ensure all materials and adhesives are compatible with the environment in which they are to be installed (e.g. corrosive atmosphere).
- g) Label Spacing: Every 20 feet for open areas, every 10 feet in congested areas. Both sides of floor and wall penetrations and changes in direction (within 2 feet of direction change).
- h) Lettering Size: As required by piping size, as indicated in **Table 2**.
- i) Label color code: As per media category, as indicated in **Table 3**. If there are specific local code requirement, sites follow as per local code. If existing, site label color code does not match **Table 3**, for expansion within the same fab, sites may follow existing color code to prevent confusion. Sites shall follow below requirement for new greenfield fab

Table 2 – Lettering Size

Pipe Size (OD)	Text Height	Label Height
1/4"	1/4"	1/2"
1/2" to 3/4"	3/8"	1/2"
1" to 2-1/2"	7/8"	1"
3" to 5-1/2"	1-7/8"	2"

Table 3 – Label Color Code

Media Category	Label Color code	
	Label Background	Label lettering
HPM Chemicals	Yellow	Black
Non-HPM Chemicals	White	Black
TMAH > 1%	Orange	Black

8.0 Design Review

Minimum three Design Reviews shall be carried out at the project site and/or at Micron office, at partial completion milestones. Acceptance and written approval of the partially completed design is required as a basis for subsequent design work. Minimum expectation requirements for each design review include:

- a) Process Development including Process Flow Diagrams and Pressure Flow calculations along with Material and Energy Consumption and Production. Preliminary equipment plans (Layout drawings), FMEA review and Safety in Design with HAZOP score according to Micron requirement.
- b) Process Design including Piping and Instrumentation Diagrams along with equipment sizing, piping sizing, component selection, and General Equipment and Piping Arrangements (Layout drawings).
- c) Detailed Design including Detailed Equipment, Piping and Support Drawings (Detailed Layout Drawings); equipment, materials, and component selection; and utility and accessory design. Final review of FMEA and Safety in Design.
- d) At different stages of BIM development.

Though the following information is related to design review, contractor shall inform Micron within the given time frame as this information is related to other disciplines.

- i. The Contractor shall indicate not later than 3 weeks after award of contract the necessity of any foundations or structural supports needed to move in equipment and tanks. Contractor shall review BIM or 3D Drawings to evaluate the move in equipment path.
- ii. Provide space management design within each of the Chemical and Solvent Rooms and areas. Meet and plan with other contractors to ensure a coordinated design including overhead lighting. Prepare drawings for Micron approval.
- iii. Review the make-up air and room exhaust system design, including distribution ducting, inside the designated areas. Based on requirements, as for some chemical rooms, the make-up air room main outlet- and exhaust inlet ducts have to be installed down to 50cm above the floor. Ensure that these locations are coordinated such that air flow is not badly obstructed.
- iv. Contractor shall study the civil work drawings, wall and floor penetration drawings, and space management drawings carefully to evaluate free height, necessary openings, access ways, common supports, etc. that are available. If there is any additional opening or supports required, contractor should include the cost for all additional requirements. All pipe routing and cable tray drawings shall be submitted within 45 days of contract award.

9.0 Approved Manufacturers

To minimize spare parts inventory, Micron may standardize equipment/component make such as pumps, motors, valves, instrumentations and etc.

Equipment/component which requires routine maintenance and replacement parts shall be "standard manufactured" items with replacement parts readily available. In addition, preference is given to equipment/component make which has local service and support.

Acceptable equipment/component make shall be as indicated in **Table 4**.

Table 4 - List Approved Manufacturers

Component Type	Approved Manufacturers
Chemical Blending Equipment	Air Liquide, Diversified Fluid Systems (DFS), Mitsubishi (MEC), Marketech (MIC), Nova Technology, Sumitomo & STI - other site approved manufacturers requires Micron approval.
High Purity PFA Tubing	Entegris, Furon, GEMU, Nichias, Parker, Pureline, Nippon Pillar, or equivalent
Clear PVC (Schedule 40) Piping	George Fisher, Hershey, Spears, Landmark Plastics or equivalent
High Purity PFA Fittings	Entegris, Furon, GEMU, Nichias, Parker, Pureline, Nippon Pillar, or equivalent
Ultra-High Purity Valves (Aqueous)	CKD, Entegris, Furon, GEMU, Parker, Pureline, Swagelok, or equivalent
Ultra-High Purity Back Pressure Regulator (Aqueous)	Parker or Equivalent
Ultra-High Purity Pressure Regulator (Aqueous)	Parker, SMC or Equivalent
Ultra-High Purity SS Valves	GEMU, FUJIKIN
Stainless Steel EP Pipe	Kaze, Valex
Chemical Pumps	Trebor (Air Operated Pumps), Iwaki (Air Operated / Centrifugal pumps), Nippon Pillar (Air Operated Pumps), Levitronix (Centrifugal pump), or equivalent
Low Purity Air Operated Sump Pumps	ARO, Yamada, Wilden or equivalent
Tank and Vessel	Applied Supreme, Edlon, Entegris, NGC, Plasticon, Sun Fluoro, Valqua, Fluoro Tech, Yodagawa or equivalent
Conductivity Sensor/Transmitter	Horiba, Burkert, Hach or equivalent with assurance for application
Flow Controller/Transmitter	Entegris or equivalent with assurance for application
Pressure Sensor/Transmitter	Entegris or equivalent with assurance for application

Component Type	Approved Manufacturers
Concentration Meter	K-PATENTS or Equivalent
Other Instruments	Micron Approved ones only or Prior approval to be obtained.
Load Cells/Transmitter	Hardy,
Titration	Semi-Chem
Programmable Logic Controllers (PLC)	Allen Bradley, GE Fanuc, Mitsubishi, Omron, Siemens, or equivalent

10.0 Testing, Commissioning and Acceptance

During the system fabrication, shipping, installation, and start-up phases, each chemical blender and distribution equipment, associated components and sub-assemblies are subjected to stringent quality checks and testing as defined in the above Section. Further the assemblies of systems and components shall be in a clean manner to maintain cleanliness during the assembly and packaging processes.

As needed all units shall be qualified both at factory and at field. Thus, the continuity shall be maintained to the extent that observations and non-conformance noted during factory test can be validated during field test. Required information with respect to FAT and SAT (or Site Testing and Commissioning) are provided in the relevant sections.

10.1 Commissioning Plan

Start-up activities are critical to the overall success of any installation. System Supplier needs to have most qualified commissioning team/field services with chemical, mechanical, control and electrical engineering capabilities. The Defect Liability Period ("DLP") of the system shall start upon the acceptance of the system.

Upon award of the contract, system supplier needs to prepare a Commissioning Plan including Certification and Validation if applicable. The Commissioning Plan contains all relevant information: administrative codes, test dates, type of test/inspection, detailed test procedure description, and specification, checklists, systems and components involved in the test, block diagrams, piping network diagrams, test equipment and calibration data, test results form, protocol form and factory test reports, certificates included. Prior to start of commissioning works, system supplier shall provide commissioning plan to Micron and only when the commissioning plan is accepted Micron, commissioning activities can be started.

The System Supplier commissioning team shall consist of a Project Manager and Commissioning Engineer(s). The team will verify the installation of the equipment, interconnecting process piping, utility piping, loop piping, instrumentation, electrical and control wiring supervision of the cleaning, start-up and qualification leading to customer acceptance of the system. Upon the End of Installation, System Supplier can start the pre-commissioning activities. During this pre-commissioning period, System

Supplier may stop all or part of the equipment or run it in various modes, with the aim of making the necessary settings and ensure it is working properly. However, if there is any damage to the chemical (Blender and Distribution) system or associated equipment or piping or associated systems/utilities, system supplier shall bear the sole responsibility without any claim from the Customer for the damage.

During this pre-commissioning period, all preliminary trials can be performed, concerning every equipment or step of treatment. These trials will enable to check the correct functioning of the mechanical, electric, electronic, thermal and hydraulic devices, according to the corresponding standards, the rules of the art and the planned conditions of functioning.

As minimum requirement, Commissioning Plan Shall Include the following information:

Activity	Date for the Activity	Required Documents	Micron's Responsibility	Remarks
Factory Acceptance Test	Date to be provided By Vendor	Refer to FAT Section	Approval of FAT Procedures and Check Lists	Documents to be submitted to Micron Minimum 3 months prior to the activity.
Installation Inspection	Date to be provided By Vendor	Refer to Installation Inspection Section	Approval of Installation Inspection Procedures and Check List	Documents to be submitted to Micron Minimum 3 months prior to the activity.
Site Acceptance Test	Date to be provided By Vendor	Refer to SAT Section	Approval of SAT Procedures and Check Lists	Documents to be submitted to Micron Minimum 3 months prior to the activity.
Site Testing, Start-up and Commissioning	Date to be provided By Vendor	Refer to Site Testing and Commissioning Section	Approval of Testing, Start-up and commissioning Procedures and Check Lists	Documents to be submitted to Micron Minimum 3 months prior to the activity.

10.2 Factory Acceptance Test

Factory Acceptance Test (FAT) is to be carried out in order to demonstrate operational and functional testing of the system. Though FAT is carried out at the factory, detailed testing of the I/O list, I/O Loop test, verification of assigned tagging both on component and on HMI, I/O firing test, field calibration of necessary instruments, instrument scaling both at HMI and SCADA, at least 2 point random check to verify the scaling of instruments, etc. Shall be performed.

All functional testing with all possible scenarios shall be carried out to confirm the sequence of operation and theory of operation. Functional check list shall be submitted before starting the performance test. Soft functional testing shall be performed in software simulation (or with simulator panel) and Actual functional checking shall be thoroughly performed at site.

All the alarms, conditions for the alarms and scenarios arising from the alarms shall be validated. Alarm priorities based on the Micron standards shall be in compliance with Micron Requirements. Alarm list comprising all alarms with different categories, priorities and follow up action to be taken upon the occurrence of the alarm shall be submitted before the starting any performance tests.

FAT is not just limited to the below statements as the tests can vary based on the chemistries used. Generally, FAT has to be performed in the presence of Micron staff and based on the circumstances, this particular FAT requirement may be changed.

- a) Contractor shall provide FAT procedure which shall include Checklist, all the tests carried out at factory prior to FAT, all the tests to be carried out during FAT, how all the tests are carried out, handling of change control forms with respect to FAT, etc.
- b) What have been approved in P&IDs shall not be compromised unless deviations granted.
- c) Component Tagging list, Equipment list (Components, Instruments and Hardware/Software information), I/O list, pressure test reports, tank lining integrity test report, factory calibration certificates, DI water grade used for flushing the system, DI Water sample analysis if available, etc. shall be provided prior to FAT.
- d) During FAT, at least the followings will be tested at the Factory.
 - i. Vendor's QA/QC procedures and QA/QC data of the system
 - ii. System size, tank sizes, mechanical integrity, accessibility to components, etc. shall be checked. System has to match the 3D modelling provided by the contractor during design stage.
 - iii. Mechanical, Pneumatic and Electrical segregation for individual engines to be checked.
 - iv. Leak test for tanks

- v. Leak test for cabinet whichever portion is necessary – cabinet bottom, sample compartments, other compartments through which chemical or UPW water or any other chemical or wastewater is passing.
 - vi. Leak test for all wet tubing/piping. Perform the same for dry tubing/piping too.
 - vii. System start up test
 - viii. EMO test
 - ix. I/O Loop test
 - x. Valve firing test
 - xi. Instrumentation scaling
 - xii. Functional Tests. If wet test cannot be performed, all functions shall be simulated either by a simulator or by a software simulation.
 - xiii. Communication with remote units/stations
 - xiv. If tank lining Integrity Test can be carried out, it shall be performed.
- e) Upon completion of FAT, Micron shall approve if the system can be sent to site.
- f) All rectification works shall be completed by contractor and shall be either retested at factory in the presence of Micron staff or at site based on the severity of the rectification work.
- g) **Appendix 1** shows a guideline for FAT.

10.3 Installation Inspection

Installation Inspection is to be carried out in order to demonstrate readiness of the system installation so that relevant Testing, Start-up and Commissioning work can be started. Thus, Site Inspection for Start-up shall precede testing and commissioning activities. The following points shall be observed prior to site inspection works. During Design of installations, some conflict can occur with installation of other disciplines according to BIM and therefore these conflicts should have been resolved before installations.

- a) Contractor shall provide Installation Inspection procedure which shall include Check List, all the tests carried out prior to/During/after Installation, how all the tests are carried out, handling of change control forms with respect to FAT, etc.
- b) All installations shall comply all relevant codes or standards.
- c) Once the Installation of Tanks, Systems, Piping work, VMBs, Power Panels, Control Panels, other field service devices, etc., all relevant reports with respect to Installation have to be check and approved by Micron.
- d) All mechanical and electrical terminations are to be checked and thus, there shall not be any open-ended pipes, tubes, ducts, cables, etc.

- e) When there are many installations proceed in parallel, relevant installation inspection can proceed one after the other based on installation completion.
- f) When there are many systems are installed, inspections and relevant documents are to be submitted for individual systems.
- g) During Installation Inspection, at least the followings checks shall be carried out.
 - i. Vendor's QA/QC procedures and QA/QC data with respect to installation.
 - ii. System size, tank sizes, mechanical integrity, Electrical Installations, etc. shall be checked.
 - iii. All piping and their integrity shall be checked
 - iv. All electrical and instrumentation cables shall be checked.
- h) Upon completion of Installation Inspection, Micron shall approve if the installations shall be used for its purpose.
- i) A guideline for Installation Inspection is provided in **Appendix 2**.

10.4 Site Acceptance Test

The Site Acceptance Test will be carried out after the completion of installation of the system at Micron facility. The purpose of the SAT is to verify that the system is supplied and installed for the system to operate according to the designed specification. At the completion of installation of the system, as stated under QA/QC, field service engineer from the contractor shall be in-charge for the SAT. Field Service engineer shall certify that the system is installed as per the requirements.

Contractor shall provide SAT procedure which shall include Checklist, all the tests to be carried out at Site prior to start-up, how all the tests are carried out, handling of change control forms with respect to SAT, etc.

If FAT was not carried out with the agreement of Micron due to some unavoidable circumstances, a thorough FAT shall be performed during SAT as per FAT guidelines provided under FAT. If FAT was done properly, some critical checks shall be performed to confirm the integrity of the system as below by a trained personal:

- a) Mechanical and electrical Integrity of the installed system.
- b) Ensuring that all the loads of the load cell-based blend tank is taken up by load cells. For this purpose, load cell calibration and linearity check shall be performed with calibrated weights. Following points shall be noted for calibration:
 - i. Based on the designed blend weight, 10 standard weights each with 1/10th of the design blend weight (approximately) shall be prepared.
 - ii. Standard weight shall be gradually loaded one by one and using the respective readings at HMI, HMI Reading Vs Standard Weight shall be established for loading.

- iii. Similarly, after the full designed blend weight is loaded, weights shall be reduced one by one and using the respective readings at HMI, HMI Reading vs Standard Weight shall be established for unloading.
 - iv. All the data points captured in the SCADA or HMI shall be attached to calibration curve for reference.
 - v. If there is any other method is established by the system supplier, it shall be given in writing and get the necessary approval.
 - vi. If the weight of any components of the blend composition (mixture) is lower than the 1/10th of the designed blend weight, that particular weight shall be taken as incremental/ decremental weight for calibration of the load cell.
 - vii. When error is calculated between HMI and actual weight, reference shall be the smallest weight. If the smallest weight is 500g and HMI shows 490g, the error % is 2%.
 - viii. When calibration is done, the highest error value during loading or unloading has to be recorded and used for error calculation.
 - ix. R^2 (coefficient of determination) for the correlation (calibration curve) shall be very close to 1.0
- c) Ensure integrity of the load cells, flow controllers, flow and pressure regulators, etc.
 - d) Ensure the completion of all the rectification works to be performed based on FAT.
 - e) Electrical and Control Panel Verification Testing: At the completion of panel installation, each Electrical and Control (Remote I/O Panel -RIO) panel to be tested by a factory-trained and certified field service engineer according to Electrical and Instrumentation & Control specification.
 - f) Confirm input voltage (120Vac or 240Vac or appropriate 3 phase supply) is within appropriate tolerance range, and that the panel indicates normal operating condition upon the 120 or 240 Vac power application.
 - g) Confirm the operation of EMOs – System, Remote panels, Room EMO and Remote EMO in TGCM room
 - h) Ensure the integrity of the PLC/HMI/SCADA programs
 - i) Ensure the availability of all the modes of operation and operability of the system different modes.
 - j) Test the Day-tank level settings and comparison with analog reading to confirm the instrumentation scaling
 - k) Ensure the system is ready for start-up

10.5 Testing, Start-Up and Commissioning

As stated under commissioning plan sections, Prior to any testing activities, Contractor has to submit the Test Plan (TP) or inspection Test Plan (ITP) for approval of Micron.

10.5.1 Testing and Start-Up

- a) Before starting any testing activities, Contractor has to inform the Owner and Purchaser (based on purchase requirement) the beginning of the tests at least one week in advance.
- b) Contractor shall produce the list for the testing and as minimum, Contractor shall perform following tests as indicated below: Based on the SAT requirement, some of the test supposed to be performed before SAT and it is the responsibility of the contractor to ensure the completion of those required tests before SAT.

Equipment	Activity	QC Point	Quality/Test Record	Responsibility of System Supplier
Tank	Pressure Test	Off-site and On-site	Pressure Test Report	Preparation for test followed by execution of the test
	Spark Test or Any other Integrity Tests for Lining	On-site	Spark Test or Any other Integrity Test report for Lining Test Report	
	Lining Thickness Test	Off-site	Test Report	
	Harness Test	Off-site	Test Report	
	Leak Test	Off-site and On-site	Test Report	
Utilities readiness before hook-up to system - UPW	Pressure Test, Metal, Particle	On-site	Pressure Test, Vacuum Test, Particle and other	Preparation for test followed by execution of the

Equipment	Activity	QC Point	Quality/Test Record	Responsibility of System Supplier
Utilities readiness before hook-up to system - PN2, GN2	Pressure, Vacuum (He Leak test), Particle.	On-site	analytical test reports	test and third-party analysis
System	FAT	Off-site	FAT Check List	Preparation of Check list and other preparations for FAT
System and all other associated equipment	Installation Inspection	On-site	Installation Inspection Check List	Preparation of Check list and other preparations for Installation Inspection
System	SAT	On-site	SAT Check List	Preparation of Check list and other preparations for SAT
Inter-connections	Pressure Test	On-site	Pressure Test Report with sections of piping	All preparation work for pressure test and execution of the test
Distribution Loop	Pressure Test	On-site	Pressure Test Report with sections of piping	All preparation work for pressure test and execution of the test
Tank	Flush with UPW followed by Metal and particle analysis	On-site	Full Metal analysis report and Particle count report	All preparation work for flushing & sampling, execution and third-party analysis
System – Through all wetted parts and take samples from - Final Discharge point	Flush with UPW followed by Metal and particle analysis	On-site	Full Metal analysis report and Particle count report	All preparation work for flushing & sampling, execution and third-party analysis

Equipment	Activity	QC Point	Quality/Test Record	Responsibility of System Supplier
Any Enclosures	Leak Test	Off-site for all On-site for some	Leak test Report	All preparation and execution
Electrical/Control / Instrumentation Test	EMO Test for all Panels	On-site	EMO Test Reports	All preparation and execution of work
	Cabling Continuity Test	On-site	Cabling Continuity Test Reports	
	Loop Test	On-site	Loop Test Reports	
	Earthing continuity check	On-site	Test Reports	
	Panels earth connections if required	On-site	Test Reports	
	Turning on Electricity to system by LEW	On-site	LEW's Power Turn-on Report	
	Any Functional test that is not performed during SAT	On-site	Functional Test Reports	
Load Cell	Calibration	On-site Calibration with Std weight Vs. HMI Readings	Calibration report with calibration curve of (Upwards and Downwards)	All preparation including volume calculations for blend tanks and execution of work
Conductivity Meter/Transmitter	Calibration	On-site	Calibration Certificate	Calibration based on relevant procedure
Density Meter/Transmitter	Calibration/Verification	On-site	Calibration/Verification Cert	Calibration/verification based on relevant procedure

Equipment	Activity	QC Point	Quality/Test Record	Responsibility of System Supplier
Concentration Meter/ Transmitter	Calibration/ Verification	On-site	Calibration/ Verification Cert	Calibration/ verification based on relevant procedure
Flow Controller	Calibration for the chemical, UPW or any other blending component to be established based on density & viscosity	On-site	Calibration Cert	Calibration based on relevant procedure
Radar Sensor/ Transmitter	Mapping and Verification	On-site	Tank Mapping information	Mapping using relevant software
Digital Level sensor settings for Day tanks	Verification of Levels and matching with Design basis	On-site	Digital Level Sensor Settings	Set the digital level sensors at correct location and fine tune the sensor sensitivity
Chemical blend Quality	Based on chemical property, Solid Content, Density, Conductivity, Chemical Concentration	On-site	Third Party Analytical Results	Sampling and third-party analysis
Verification of online metrology	Based on chemical property Density, Conductivity, Chemical Concentration	On-site	Analytical Data and corresponding Field instrument values and SCADA Data	Verification of field SCADA values with lab analysis
Sump pump Enclosure	Leak test	On-site	Test Report	Preparation and execution
Sump	Leak test	On-site	Test Report	Preparation and execution

Note:

- i. As for all the test, Micron shall witness, approve, and accept. Without witness, approval and acceptance of Micron, commissioning data shall not be accepted
 - ii. Some of the line items are common for many sections or equipment and therefore, contractor's checklist has to be with individual ones so that testing will be done individually, and reports can be generated accordingly.
 - iii. Some of the above tests are to be performed after start-up and therefore, contractor shall perform those tests after start-up.
 - iv. Load cell calibration and linearity check methodology is explained in previous section.
 - v. When conductivity cell calibration is performed, conductivity cell shall not be removed from the measurement location. Liquid (blended sample also can be used) with Known conductivity value shall be filled in the required section of the instrument line and calibration shall be performed. Zero calibration can be performed without any content in the required line. Zero Calibration shall be performed first.
- c) Once all the required test for start-up is completed, system will be ready for start-up and start-up shall be performed based on start-up procedure submitted by the contractor. Though start-up procedure is available, a start-up check list to be submitted by contractor prior to start-up to confirm and system shall be checked based on start-up checklist. As minimum, start-up check list shall include the following points:
- i. All the required safety checks
 - ii. Approval of the start-up and shutdown procedures by Micron
 - iii. System installation/construction acceptance from Micron
 - iv. Start-up can be initially performed with UPW only. All the wet tests to be performed based on the chemical properties of chemistry shall be listed.
 - v. Availability of all required medias (slurries, chemicals, etc.)
 - vi. Availability of final draft of Operation and maintenance Manual
- d) Based on the checking performed in accordance to start-up check list, the readiness for start-up will be confirmed.

- e) Based on the chemistry criteria, system start-up shall be performed using UPW. If UPW cannot be used, correct chemistry shall be used for start-up.
- f) Once start-up is performed according to start-up procedure, system shut-down shall be checked in accordance to shut-down procedure. Thus, start-up and shutdown is completed.

10.5.2 Commissioning

- a) Based on the given schedule, Contractor shall inform Micron the beginning of the commissioning at least two weeks in advance.
- b) Commissioning of all components and the total system shall be witnessed by the Micron.
- c) If FAT or SAT was performed thoroughly, selected critical components such as pumps shall be commissioned based on equipment commissioning guideline provided by the equipment manufacturer. Commissioning checklist for the critical components shall be provided by the contractor.
- d) Contractor shall also provide the commissioning checklist for the system.
- e) All the system parameter settings set during commissioning shall be approved by Micron. Until final agreement is achieved between contractor and Micron, contractor is liable for the parameter settings set on the system HMI/SCADA.
- f) Instrument and equipment accuracies shall be confirmed during commissioning activities
- g) When conductivity or any other parameter is used to represent chemical concentration, calibration curve of Conductivity (or other relevant parameter) versus concentration shall be established for the desired concentration range. This curve shall be established using the actual chemical(s) used for blending.
- h) If conductivity (or any other parameter) is selected for representing concentration, as conductivity is a temperature dependent parameter, temperature compensation coefficient shall be found for the required blend. This temperature compensation coefficient shall be used for temperature compensation coefficient and shall be used appropriately in the PLC programing. It shall be noted that cell constant is different from temperature compensation coefficient.
- i) All the functional checks shall be carried out during commissioning
- j) As minimum, following activities as shown below shall be performed during commissioning.

Equipment	Activity	QC Point	Quality/Test Record	Responsibility of System Supplier	Responsibility of Micron
Pumps and other critical equipment	Commissioning based on equipment manufacturer's guideline.	On-site	Commissioning report	Performing commissioning for the respective equipment	Witness, approval and acceptance.
Instruments	Scaling between field and Remote (HMI and SCADA)	On-site	Instrument Scaling Reports	Preparation and execution	Witness, approval and acceptance.
Metrology Instruments	Accuracy of the instruments for the particular chemistry	On-site	Verification Reports	Performing verifications	Witness, approval and acceptance.
System	Functional Test comprising all possible scenarios	On-site	Functional Test Reports	Preparation of Functional Test checklist and performing functional test	Witness, approval and acceptance.
Sump Pump	Pumping Rates	On-site	Pump Capacity Test Report	Preparation and execution	Witness, approval and acceptance.
Sump Pump	Functional Checks	On-site	Test Reports	Preparation and execution	Witness, approval and acceptance.
System	Verification of Alarms – This also can be covered under functional test	On-site	System alarm and SCADA alarm reports	Preparation and execution	Witness, approval and acceptance.
Blended Chemistry	Qualification accuracy, repeatability and reproducibility	On-site	Qualification Test Reports	Preparation of blends in automatic mode, sampling and analysis	Witness, approval and acceptance.
System	Leach test based on chemistry	On-Site	Leach Test Report	Preparation and execution	Acceptance

Equipment	Activity	QC Point	Quality/Test Record	Responsibility of System Supplier	Responsibility of Micron
System	Simulation for Power Failure test	On-site	Power failure simulation test report	Preparation and execution	Coordination, witness, approval, and acceptance
System	Simulation of power failure from one source to ensure seamless take over.	On-site	Power failure simulation test report	Preparation and execution	Coordination, witness, approval, and acceptance
SCADA	Pages, Process Flow Diagrams, Trends, SPC if included,	On-site	Printed pages, PFDs, Trends, SPC information	Preparation and work with SCADA Vendor depending on scope	Coordination, witness, approval, and acceptance
SCADA	Alarm Testing, Event Tracking, Scaling, Prioritization, etc.	On-site	Generated Alarm list, Event list, Scaling information, priority setting, etc.	Preparation and work with SCADA Vendor depending on scope	Coordination, witness, approval, and acceptance

- k) Qualification procedure needs to be submitted by contractor and to be approved by Micron. Sampling points, how many samples to be taken for accuracy and repeatability & reproducibility. Measurement detection limits and percentage of error, etc. shall be specified in qualification procedure.
- l) Startup and validate System in compliance with requirements of the contract documents. Provide third party analytical testing as necessary to verify proper performance for System qualification and acceptance. The Chemical blender and distribution System shall continue demonstrating adequate performance under full load conditions.
- m) Contractor shall submit requirement of Chemical/ Chemical volumes for commissioning. Chemicals and chemical logistics, including chemicals required for startup are the responsibility of MICRON
- n) Simulate power failure and Confirm no loss of functionality occurs. Test and confirm the supply from UPS or battery for specific components.

- o) Simulate power failure from one source and ensure the operation of the system without any impact. For both sources this test to be performed.
- p) Commissioning is deemed complete with following document to be proven.
- q) System Commissioning Completion Certificate signed-off by Micron.
- r) During the commissioning the training of the future operation team shall be conducted.
- s) If there is any additional requirement specified in other disciplines for commissioning of the respective disciplines such as SCADA, Electrical and Instrumentation, refer the respective discipline document.

10.6 Performance Test

Performance test of the system shall be performed before the final acceptance of the system. This includes running of the complete system – transfer, blending and dispense based on the design concept used - at the required flow without any interruption or any major changes in the system installation or system control units for at least continuously 14 days (guarantee parameters in specification).

The test is considered failed if:

- a) Failure by a self-cause and/or substantial intervention at the system, to ensure function.
- b) Specification parameter not achieved.
- c) Safety devices not responding on failure.

Upon failure, re-start of the tests is expected, and the costs shall be fully borne by the Contractor.

Performance test can only be performed only after the following actions

- a) Acceptance of qualification reports
- b) Acceptance of all other commissioning reports
- c) Settings of all agreed parameters on HMI/SCADA

When the performance test is in progress, Micron shall request to pull samples and analyze to verify the blending accuracy and repeatability & reproducibility of the system. Contractor also shall submit the sampling plan during the performance test period.

Performance test will be accepted upon submission of performance test report in addition to all required site recordings. Performance test report shall comprise the followings:

- a) Blender application summary – This is a table with all basic recipe calculations, flow rates, consumption rates, etc.
- b) Blender performance trend (conductivity or concentration trend during qualification or transfer depending on the type of blender) for individual batches (for batch blenders) or for lowest/highest volume fill to the day tank at lower and higher blend rates (for in-line blenders)
- c) If conductivity is used as the reference parameter, Conductivity Vs temperature for a blend closer to the required recipe. This shall be conducted in controlled environment at different temperatures such as 19°C, 21°C, 23°C, 25°C and 27°C. The equation obtained from the graph shall be used for temperature compensation.
- d) Temperature trend during transfer
- e) Table comprising online conductivity, lab conductivity and lab concentration for different target online conductivities (for in-line blender) or target recipes (for batch blender). Develop useful graphs to explain the correlations.
- f) Once target conductivity or recipe versus concentration is developed, target conductivities for the required upper, lower, and middle specifications (or control limits based on what is required) shall be predicted for the performance of the system.
- g) Actual Performance Trends and tables be based on the predicted Targets.
 - i. System performance trends.
 - 1. Trends to prove the qualification period (has to be less than 1 minute for in-line blenders).
 - 2. Trend of reference parameter (such as conductivity) with specified bandwidth of the target value.
 - 3. Maximum acceptable time duration at a time for out of bandwidth (control and specification limits) e.g. for 4 sec. within which the system supposed to operate.
 - 4. Maximum time duration within given time period for out of bandwidth (control and specification limits) e.g. for 7 sec. in a minute.
 - 5. After qualification, total time out of control limit shall be calculated and shown in the performance trend.
 - 6. Requirement for load cell or flow controller taring or zeroing after each blending cycle.
 - 7. Requirement for flushing after each blending cycle.
 - ii. Sample analysis results
 - 1. Samples shall be taken for accuracy, repeatability and reproducibility as shown below. S1-9 refers to Sample 1 - 9. Blenders

For Batch Blenders

Sample	Lower Control or Lower Spec. Limit	Control Limit	Upper Control or Upper Spec. Limit
For Accuracy	Fresh sample after stabilization – S4	Fresh sample after stabilization – S1	Fresh sample after stabilization – S7
For Repeatability	Fresh sample for new blend prepared – S5	Fresh sample for new blend prepared – S2	Fresh sample for new blend prepared – S8
For Reproducibility	Fresh sample for new blend after shut-down and start-up the system – S6	Fresh sample for new blend after shut-down and start-up the system – S3	Fresh sample for new blend after shut-down and start-up the system – S9

- Note:** (i) For each tank 9 samples shall be taken as prescribed above
- (ii) If there are 3 blend tanks, 27 samples to be taken for qualification. However, if the number of samples can be reduced based on the approvals granted by Micron system engineer.

For in-line Blenders

Sample	Lower Control or Lower Spec. Limit	Control Limit	Upper Control or Upper Spec. Limit
For Accuracy	Fresh sample after stabilization - S4	Fresh sample after stabilization - S1	Fresh sample after stabilization – S7
For Repeatability	Fresh sample after stop and start the system for new blend – S5	Fresh sample after stop and start the system for new blend – S2	Fresh sample after stop and start the system for new blend – S8
For Reproducibility	Fresh sample for new blend after shut-down and start-up the system – S6	Fresh sample for new blend after shut-down and start-up the system – S3	Fresh sample for new blend after shut-down and start-up the system – S9

- Note:** (i) For each tank 9 samples shall be taken as prescribed above
- (ii) If there are 2 blend lines with separate flow controller and in-line blenders, 18 samples to be taken for qualification.
1. Acceptable RE% shall be specified (or 1% if not mentioned) for all the samples.
 2. If one of the 9 samples fail, the specific tank or specific in-line blender to be qualified again.
 3. If there is any change in programming between qualifications of tanks or in-line blenders, sampling shall be performed again.
 4. Parameters to be measured for any sample shall be agreed by all parties. However, concentration, conductivity (lab and on-line), and any other critical parameter (or corelated parameter) shall be measured for qualification.
 5. All the sample results with lab reports shall be attached in the performance report to validate the performance results.
 6. All the sample results shall be used to corelate and to ensure the performance of the blender.
- (iii) If any other points are discussed during design review, those shall be included in the final design.

10.7 Acceptance and Certification

- a) Equipment vendors performance report shall be accepted by Micron which is a prerequisite for acceptance and certification.
- b) Certification shall be done by the System Supplier together with the Micron or appointed TGCM contractor
- c) The purchaser may request a third party to be involved in the certification phase. This will be at no additional cost to the system supplier.
- d) The start of the certification measurements and tests shall be announced at least two weeks in advance. An introductory meeting and a presentation of the certification set-up shall be organized by the System Supplier for all involved parties.
- e) For the certification and acceptance criteria for the system shall have the followings as Minimum:
 - i. Meet all requirements on EHS
 - ii. Meet all installation requirements
 - iii. Meet all required technical specifications
 - iv. Completion of all set points with logical reasons – System Parameter Settings with approval from system engineer
 - v. No critical alarms for 14 days
 - vi. Submission of all relevant documentation as stated under section submission of documents.
- f) Acceptance is deemed to be the complete specified and fully functional delivery of the works (System) to the Owner/Purchaser after testing, complete with service requirements e.g.
 - i. Completed Quality Assurance Program (QAP) report.
 - ii. Completion of the as-built drawings of the complete installation.
 - iii. Documentation of operational and functional acceptance tests, and of third-party laboratory analysis.
 - iv. System has been validated and performance satisfactorily demonstrated.
 - v. Completion of the operation and maintenance manuals
 - vi. Completion of the warranty certificate(s)
 - vii. Completion of punch list items.
 - viii. Completion of training programs.
 - ix. Any other points stated in the contract. An official report (System Final Acceptance Certificate) of this successful acceptance shall be prepared by the Contractor and signed off by Micron.

- g) If during operation of the system, it is determined that the systems/sub-systems do not function properly or do not comply with the specified requirements of the contract, a renewed testing and acceptance procedure cycle shall take place.
- h) Material, components, equipment, assemblies, and subsystems will be considered defective and replaced if unable to pass tests and inspections in conformance with contract requirements.
- i) With respect to the rework, all associated costs shall be borne entirely by the Contractor, including the fees and transportation / accommodation costs of the Owner/ Purchaser's representative(s).

11.0 Quality Management

11.1 Quality Control Requirements

QA/QC is an indispensable part of high purity system manufacturing, installation, testing and commissioning. The following points with respect to QA/QC shall be observed during the project.

- a) Provide skilled installation personnel who are certified in their trade.
- b) Provide job training and instruction of pipe joining techniques for personnel assigned to such duties. Welders certificate to be submitted as part of the submittal process. Clear PVC bonders need to be certified onsite.
- c) Include training and certification by pipe manufacturer for joining technique selected.
- d) It is the scope of the contractor to co-ordinate with Micron to schedule and conduct source inspections at any manufacturing facility or assembly plant prior to shipping any equipment designated for this project.
- e) Performing appropriate inspection and testing is necessary for all equipment and provide inspection report prior to source inspection by Micron.
- f) Maintaining records is very important and Micron will audit the documents when necessary.
- g) In order to control and assure quality of all works related to the chemical (Blender and Distribution) system, establish a quality assurance program (QAP) which shall be maintained throughout the project by Contractor. Clearly document QAP in the Bid. QAP is subject to monitoring during the course of project execution. QAP may include:
 - i. Material handling
 - ii. Material acceptance or rejection criteria
 - iii. Replacement of material when it is rejected.
 - iv. Incoming Quality Checks - material and equipment inspections.
 - v. Skill level of Thermoplastic (non-clear PVC / PFA) Pipe fitter / Welder and trainings certificates.
 - vi. Skill level of clear PVC & PFA Pipe fitter / Bonder and training certificates.

- vii. Skill level of Stainless-Steel Pipe fitter / Welder and training certificates.
 - viii. Filed performance and credentials of these skilled workers
 - ix. Fabrication procedures and clean room protocols
 - x. Thermoplastic and Stainless-Steel high purity pipe welding procedures
 - xi. Design compliance checks.
 - xii. Monitoring of submittals provided, and incorporation of comments received.
 - xiii. Development of the detailed System startup and validation plan.
 - xiv. Coordination between Contractor and Micron.
 - xv. Coordination between the Contractor and site installation activities. Coordination of multiple disciplines within the Contractor's organization (i.e. electrical, I&C, shop fabrication, mechanical).
 - xvi. Compliance on all necessary tests that are performed on-site and off-site
 - xvii. Verification the test reports are generated in accordance with the project requirements, including appropriate signatures.
 - xviii. Verification of Licenses used for PLC programming and HMI/SCADA development
 - xix. Certifications and credentials of PLC programmer and SCADA developer
-
- h) Micron reserves the right to review facilities or fabrications in process at any time or finished equipment prior to shipping, and to witness shop testing.
 - i) Contractor shall assign an individual who will be solely responsible for quality control functions, resident on job site. Quality control functions to include as minimum; cleanliness checks, component inspections, welding inspections per SEMI F81-1103, conformance to F78-0304 installation practices, bonding inspections, craftsmanship inspections.
 - j) Workmen shall be thoroughly experienced in their trade and fully capable of manufacturing and installing the assigned works. All works shall be performed in accordance with the best practice of the trade, as recommended by the manufacturer and/or as directed and approved by the Micron. Any work that is determined to be of substandard or poor quality by the Micron shall be removed and reinstalled at no additional cost to the Contract. All costs resulting from any schedule impact generated by such actions shall be fully borne by the Contractor.
 - k) An experienced supervisor shall be continuously in attendance at the jobsite during all phases of construction to co-ordinate and supervise the works.
 - l) All workers shall follow the site safety rules, regulations and practices imposed by Micron.

- m) Based on the size of the project, Micron shall request a QA/QC engineer during site construction.
- n) All QA/QC reports shall be submitted weekly during construction period. The submittals shall be in accordance with latest Micron practice.
- o) Factory test reports from manufacturer prior material/equipment delivery are to be submitted to warrant that the delivered material/equipment meet the required specification.
- p) The full authority of the contractor management shall be given to a member of their full-time site staff. This employee shall have the authority to commit financial and technical decisions of a magnitude associated with the size of the contractor's works' scope.
- q) Based on site conditions and progress, Micron has the right to request for additional QC and/or certification services which can be provided partly or fully by a third party during the installation of the systems and in regular intervals.
- r) Engage a qualified testing agency to perform tests and inspections that is outside the core competency of the Contractor.
 - i. Perform tests and inspections in conformance with equipment manufacturer recommendations.
 - ii. Prepare test and inspection reports.
- s) It is contractor's responsibility to provide factory trained field service engineer or engineers to perform System checkout and startup. Engineers who performed Factory FAT/SAT test prior to shipment are preferred to be the field service engineer. Field service engineer needs to certify in writing that the supplied equipment has been installed correctly with proper piping and wiring connections.

11.2 Design Validation

Complete design including process flow diagrams, P&IDs, equipment & pipe specification, pipe and equipment plans and details, equipment skid general arrangements, storage vessel fabrication drawings, equipment support and restraint details, and pipe support and restraint details along with the referenced specifications, shall be validated prior to start of any other work in compliance with Micron requirements. All the source calculations are to be submitted to support the design review. HAZOP and FMEA shall be the performed in detail such that safety and robustness of the system shall be maintained throughout the life cycle of the system.

3D modelling drawing validated to make sure of the requirements on system segregation, component accessibility, space management within the system and operation & maintenance. NPSH shall be validated through modelling such as Computational Fluid Dynamics (CFD) software. Hydraulic modeling to validate flow requirement shall be performed with AFT Fathom 7.0 or equivalent software. BIM shall be submitted in different phases in order to evaluate the piping and system layout and space management.

Sequence of operation (SOO) shall be submitted, reviewed and validated. Sequence of operation is the basis for safe, risk free and robust operation. Based on SOO, functional tests and alarm list shall be prepared and validated.

HAZOP and FMEA shall be submitted and reviewed. Based on HAZOP and FMEA observations, SOO followed by Functional test and alarm list shall be updated and validated.

In addition to the general design validation requirements, there are specific design validation requirements for chemical blender systems and following areas of design for chemical blender systems will be validated:

- a. Methodology of Blending for the specific final product (Batch blender or In-line blender). Suitability of the blender for possible quality and quantity of the blend.
- b. Quality measuring (conductivity, concentration) and quality control (flow controller, load cell) instrumentation used in the chemical blender.
 - i. Integrity of the instrument including material of construction
 - ii. Measurement accuracy of the instruments
 - iii. Measuring range and required range
 - iv. Usual life span
 - v. Required frequency for calibration
 - vi. Past experience on drifting for such blend
 - vii. Effective use of factors such as cell constant, temperature compensation factor,
- c. Selection of mixing methodology
- d. Time required for stabilization of the blend
- e. The required operational control and specification range for direct or indirect parameters (e.g. concentration can be direct parameter and conductivity can be indirect parameter) that are measured online.
- f. What is the acceptable frequency for out of the control and specification limits?
- g. Setpoint for maximum acceptable time duration at a time for out of the bandwidth (control and specification limits) e.g. for 4 sec.
- h. Setpoint Maximum acceptable time duration within given time period for out of the bandwidth (control and specification limits) e.g. for 7 sec. in a minute.
- i. Acceptable conflict % between controller (flow controller) set point and process value which will be based on feedback (concentration or conductivity) value.
- j. Blend quality RE%: Decision on the control limits and specification limits and the appropriate usage of control limit and specification limit. It will be critical to ensure which one (control or specification limit) will take precedence for RE % calculation.

12.0 Training Program

Training Micron staff or the appointed TGCM contractor is very essential to ensure safe and consistent operation of the systems. The actual training dates shall be agreed between Micron and system supplier. System supplier shall submit a detail training program for approval 4 weeks before training commence

- a) Classroom and In-the-field training shall be provided in Micron's premises
- b) As explained below the training shall be conducted in three categories.
- c) If special equipment or instruments used in the system, equipment / instrument manufacturer's personnel shall be brought in for instruction on that particular equipment/instrument.
- d) As operation teams are working on shift, training is required for two teams.

12.1 Training for System Operation

As a minimum system operation training shall include:

- a) Basic P&ID
- b) Basic theories of the system and sequence of operation
- c) Properties of the chemicals
- d) Operation of the system using Human Machine Interface
- e) Modes of operation and when/how to be effected.
- f) Critical/warning/other alarms and appropriate measures to ensure proper operation of the system.
- g) Critical observations for the components of the system
- h) Basic troubleshooting of the system
- i) Chemical preparation (if applicable)
- j) Sample collection
- k) Explain how to guarantee accuracy of online & batch blending and dilution. Discuss the properties of the blended slurries (for example: conductivity curves) and the of role of the metrology analyzers

12.2 Training for System and Equipment Maintenance

As the group of team members attending this training are from service team, this training shall be conducted to address the service issues. As a minimum, system and equipment maintenance training shall include:

- a) All major components of the system and their operation and maintenance
- b) Troubleshooting of major components such as pumps, VFDs, Instruments, etc.
- c) Control system and basic PLC training to troubleshoot control issues
- d) Electrical power system training to troubleshoot electrical related issues
- e) Instrument calibration/verification/validation to ensure the proper cooperation of the instruments.

- f) Explanation on Critical/warning/other alarms and appropriate measures to ensure proper operation of the system.

12.3 Training for System Engineers

As the group of team members attending this training are engineers, this training shall be conducted to ensure the operation and maintenance of the system without much support from system supplier. As a minimum, training for system engineers shall include all the points in detail requested in the above sections and detailed system troubleshooting methodologies.

13.0 Documentation

Documentation is very important to carry out and complete the project very effectively. Submission of the required documents at the right time moves the project as per the project plan. **Appendix 6** shows the summary of documents to be submitted at different phases of the project by the vendor. The following list below shows some critical document details that shall be submitted at the different phases of the project:

13.1 Calculation Documents

When components or process parameters in the systems are sized or rated, these have to be proved with calculations. Thus, all the relevant calculations shall be submitted to Micron for review and approval. System supplier shall bear all responsibilities for any error in the calculations and thus bear whole responsibility to ensure the system operation within the required specification. As Minimum, the following calculations shall be provided by system supplier.

- a) System Capacity
- b) Tank Sizing
- c) Pump Sizing
- d) Dampener Sizing
- e) Pipe Sizing for suction, discharge and distribution
- f) Pipe sizing for all Interfaces
- g) Pipe sizing for Drain line and Drain Headers
- h) Filter Sizing
- i) Heat Exchanger Sizing
- j) In-line Blender sizing
- k) Mixer Sizing
- l) Minimum Recirculation flowrate
- m) Time durations for different stages
- n) System and Loop pressure loss calculation
- o) Exhaust Calculations
- p) Instrument Air consumption calculations
- q) CDA consumption calculations if CDA is used.
- r) Any other required System or Equipment calculations

13.2 Sequence of Operation (SOO) Document

Sequence of operation of a system will have to explain in detail how the system operates to meet the required specifications of the system and how the operators and engineers shall intervene to ensure smooth operation of the system if the system goes into alarm mode. When sequence of operation is presented by System Supplier, the following points shall be considered:

- a) SOO can be presented in paragraph format or point format or flowchart format.
- b) If the complete system is supplied as two separate modules such as transfer unit and dispense unit, etc., while SOO for each unit is presented, all interfaces and impacts due to operation of separate modules to be clearly discussed in SOO.
- c) The sequence from Drum hook-up (or even Bulk Transfer) to dispense shall be discussed in SOO.
- d) All different stages of operation such as rinsing, qualification, stand-by, switching, etc. shall be discussed in detail.
- e) SOO for different modes of operation shall be discussed
- f) All scenarios that can occur during operation shall be considered in SOO
- g) All alarms that will be generated during operation of the system shall be linked to SOO
- h) The reasons for alarm status and required actions to be taken by operators and engineers are to be explained.
- i) Limitations of the automation and control available in the system shall be clearly discussed so that the operators and engineers would know the requirement of their actions for the smooth operation of the system.

13.3 Software Modeling Documents

Software models become inseparable from design of any system. Software designs shall be available not only for design of the system but also for space optimization and space management.

13.3.1 Building Information Modelling

BIM for Rooms, Sub-Fab Area, Common Corridors, Etc. BIM shall be used to get all intelligent engineering information about the facilities prior to construction.

13.3.2 3-Dimensional Modelling

3D Modelling for Systems to get all intelligent engineering information about the system prior to construction.

13.3.3 Computation Fluid Dynamics or Equivalent

CFD or Equivalent shall be used to identify the hot spots for in-line or batch blenders. While CFD can be used tank levels and pump suctions, length of dip-tube, transfer of return bubbles

into pump suction, etc., it can be used for predicting homogeneity of the product of blenders. Time duration for blending (either stirrer or recirculation) can be properly predicted for batch blenders. Flow rate ranges and time for stabilization can also be properly predicted using CFD. All relevant hot spots shall be discovered so that the system shall be designed for better performance.

13.3.4 Fluid Dynamic Simulations for Network Pressure Drop and Flow Distributions

As the tool consumption is dynamic, system blended chemical supply shall be equivalently dynamic to maintain system supply specifications. This kind of software shall be used to ensure system capabilities to deliver blended chemical within the system specifications at different scenarios. Software such as AFT Fathom or PIPEFLO can be used for this application. Network pressure of chemical, UPW and (or) other constituents is very important inline blenders to produce consistent blend composition.

13.4 Equipment List Document

Equipment List usually consists the list of major equipment and other major components of the systems along with the information such as Equipment Name, Manufacturer/Brand, Model, Part number, serial number, contact information for future purchasing (Principal and Local Agent), etc. Equipment list is to be used as an easier aid not only for future purchasing of equipment or spare parts but also to know the services rendered by the equipment supplier. Continuous improvements and issues related to the equipment and instruments can be known from the supplier. Operation and Maintenance Manual

Operation and Maintenance manual is not just the compilation of different equipment and instrument manuals. O&M manual shall be very detail explaining about the design, configuration, operation, and control philosophy of the system that includes process, control network, and control and other components.

13.5 Summary List of Final Project Hand-over Documents

Below is the minimum acceptable requirement for final project hand-over documentation.

- a) Contract documents
- b) Specifications: Mark and record each Product Section description of actual products installed with the following details:
 - Manufacturer's name and product model/number.
 - Product substitutions or alternates used.
 - Changes made by Addenda and Modifications.

c) Mechanical shop drawings:

- Room layout drawings
- VMB/MHB/MHEB Schematics (inclusive Equipment P&ID, Distribution P&ID, Utilities P&ID)
- Piping layout drawings (Chemical rooms, Sub-fab, CMP areas, Maintenance rooms)
- Section drawings of pipe layout
- Auxiliary equipment (such as dip tube change cabinets, drum cabinets, sampling station etc.)
- Sump pump system schematics
- Life support system (such as CO2 Fire suppression system etc.)
- Bill of Material for components

d) Electrical Shop Drawings

- Chemical (Blender and Distribution) tool interface drawing
- Overall single line drawing
- Single line drawing for individual panel
- Panel drawings
- Panel location drawings
- Overall control architecture drawing
- Individual IO panel schematic and location drawing
- Database or I/O allocation matrix
- I/O schematic
- EMO schematic
- Electrical circuitry drawing for all equipment
- Bill of Material for components

e) Engineering Calculations

- Utilities flow and pressure calculations.
- Electrical load calculations
- System flow and pressure calculations
- R&R data from equipment manufacturer for slurry and blend systems
- Design matrix

f) Product Data

- Technical specifications of systems
- Cut sheets for all major components
- Spare parts list

g) Test Certificates

- Pressure decay test reports
- Liquid particle count test report
- PN2 particle count test report
- Metallic test report
- Assay R&R test report
- Controls Checklist
- Safety Checklist
- Functional check list of CDS
- Functional check list of SCADA
- Functional checklist of SCADA/CDS interface
- Functional checklist for all auxiliary equipment
- Field I/O and Solenoid Panels Test Report

h) Operation and Maintenance Manuals for all installed equipment

i) Softcopy of all "As-built" drawings and documentations.

j) Softcopy of source code for all PLC programs (if contractually agreed).

k) Certificate of Warranty(s)

14.0 Reference Specifications:

14.1 Codes and Standards

Chemical Blending systems and sub-systems shall be designed according to IBC, IFC, NEC, NFPA and other applicable local codes and regulations and/or adopted equivalent standards. The following Guidelines and Codes in **Table 5** below shall be applied appropriately.

If the local code requirements are stricter than the any of the codes/standards mentioned in the table below, compliance to the requirements of the local code/standard shall be met by the contractor. Otherwise compliance to the international/semiconductor codes/standards shall be met as stated above.

Table 5 – Reference Specification: Codes & standards

S/N	Standard/Code		Description
1	FM	7-7/17-12	Semiconductor Fabrication Facilities
2	IBC	2015	International Building Code (IBC)
3	ICCEC	2006	ICC Electrical Code (ICCEC)
4	IFC	2015	International Fire Code (IFC)
5	IMC	2006	International Mechanical Code (IMC)
6	NEC		National Electric Code
7	NFPA	30	Flammable and Combustible Liquids Code
8	NFPA	55	Compressed Gases and Cryogenic Fluids Code
9	NFPA	70 – 2005	National Electrical Code (NEC)
10	NFPA	110 – 2002	Emergency and Standby Power Systems
11	NFPA	318 – 2012	Standard for Protection of Semiconductor Fabrication Facilities
12	NFPA	704 – 2001	Identification of the Hazards of Materials for Emergency Response
13	OSHA	29 CFR 1910	United States Occupational Safety and Health Administration (OSHA)
14	P&FM	2005	Import of Petroleum & Flammable Materials (P&FM: Fire Safety Regulations)
15	SEMI	E49.2	Guideline for the qualification of polymer assemblies used in ultra-pure water and liquid chemical systems in semiconductor process equipment
16	SEMI	E49.3	Guide for Ultra High Purity De-Ionized Water and Chemical Distribution Systems in Semiconductor Manufacturing Equipment
17	SEMI	E49.4	Guide for High Purity Solvent Distribution Systems in Semiconductor Manufacturing Equipment

S/N	Standard/Code		Description
18	SEMI	E49.5	Guide for Ultra High Purity Solvent Distribution Systems in Semiconductor Manufacturing Equipment
19	SEMI	E49.6	Guide for Subsystem Assembly and Testing Procedures – Stainless Steel Systems
20	SEMI	E49.7	Purity Guide for the Design and Manufacture of Ultrapure Water and Liquid Chemical Systems in Semiconductor Process Equipment
21	SEMI	F1	Specification for Leak Integrity of High-Purity Gas Piping Systems and Components
22	SEMI	F2	Specification for 316L Stainless Steel Tubing for General Purpose Semiconductor Manufacturing Applications
23	SEMI	F3	Guide for Welding Stainless Steel Tubing for Semiconductor Manufacturing Applications
24	SEMI	F31	Guide for Bulk Chemical Distribution Systems
25	SEMI	F34	Guide for Liquid Chemical Pipe Labeling
26	SEMI	F39	Guide for Guideline for Chemical Blending Systems.
27	SEMI	F41	Guide for Qualification of a Bulk Chemical Distribution System used in Semiconductor Processing
28	SEMI	S2	Safety Guidelines for Semiconductor Manufacturing Equipment
29	SEMI	S6	Safety Guidelines for Ventilation
30	SEMI	S7	Safety Guidelines for Environmental Safety and Health (ESH) Evaluation of Semiconductor Manufacturing Equipment
31	SEMI	S8	Safety Guidelines for Ergonomics/Human Factors Engineering of Semiconductor Manufacturing Equipment
32	SEMI	S10	Safety Guidelines for Risk Assessment
33	SEMI	E33	Specification for Semiconductor Manufacturing Facility Electromagnetic Compatibility
34	SS	532: 2007	Code of Practice for the Storage of Flammable Liquids (Singapore Standard)
35	UFC	-	Uniform Fire Code (UFC) (especially articles 51,79 & 80)
36			All other references attached within the Micron Documents – Electrical, Controls, and Safety.

The table above specifies relevant codes and standards for different disciplines. Some specific ones related to chemical and solvent are listed below:

- a) Exhausting a Cabinet
 - i. Exhaust and Ventilation - Semi S2 (Section 22)
 - ii. Design Principles Test Methodologies for Equipment Exhaust – Appendix 2 of Semi S2
- b) Fire Suppression
 - i. Fire Protection - Semi S2 (Section 14)
 - ii. Fire Risk Assessment and Mitigation - Semi S14
- c) Labels, Warning Indications
 - i. Electrical warning labels – Relevant section in EN61010
 - ii. Hazard Warning Labels and Hazards Labels – relevant sections in Semi S2 and S1
 - iii. IEC 61310 Safety of Machinery – Indication, Marking and Actuation.
- d) Pressure Vessels (15 psi and above)
 - i. PED - EC Pressure Equipment Directive
 - ii. ASME B31.3 Process Piping - B31.3 Chapter X High Purity Piping
- e) Safety Interlocks
 - i. Semi S2 (Section 11, Safety Interlock Systems)
 - ii. Semi S2 (Section 12, Emergency Shutdown)
 - iii. Semi S2 (Section 17, Hazardous Energy Isolation)
- f) Seismic Protection – this is generally covered under I & C
 - i. Semi S2 (Section 19, Seismic Protection)
 - ii. Semi S2 (Related Information 5, Seismic Protection)
- g) Solvent systems
 - i. ATEX European standard for hazardous equipment
 - ii. UL 1604 Electrical Equipment for use in Class I and II, Hazardous Locations
 - iii. UL 913 Intrinsically Safe Apparatus... for use in Class I, II, III ... Hazardous Locations
 - iv. UL 698 Industrial Control Equipment for use in Hazardous Locations
 - v. NFPA 30 - Flammable and Combustible Liquids Code
- h) Solvent Tanks
 - i. UL 2085 Tanks for Flammable and Combustible Liquids
 - ii. HP & UHP Stainless Steel Solvent Distribution Installations
- i) SEMI F78-0304: Practice for Gas Tungsten Arc (GTA) Welding of Fluid Distribution Systems in Semiconductor Manufacturing Applications.
- j) SEMI F81-1103: Specification for Visual Inspection and Acceptance of Gas Tungsten ARC (GTA) Welds in Fluid Distribution Systems in semiconductor Manufacturing Applications.

15.0 Document Control

15.1 Approval

GLOBAL_FAC_GFTT_APPR

15.2 Notification

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_NOTIFY
FCG_F2_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

15.3 Final Viewer Access

All Micron sites Facilities Team Members

15.4 Retention

Adherence to the requirements specified in the
[Global Facilities - Document Control Procedures](#)

15.5 Review

Biennially (two years)

16.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	03/31/21
1	6.0	Added 6.2.2 item u	03/29/23

APPENDIX**Appendix 1 - Guideline for FAT**

Name of Inspector:	FAT Date:
Name of Organization:	Person in Charge from Vendor:
Name of Fab:	Name of Vendor/Supplier:
Name of System:	Job/PO Number:

S/N	Items for Inspection	Verification	Satisfactory Remarks
1.0	Readiness of vendor with FAT checklist		
2.0	Availability of approved P&IDs		
3.0	Availability of single line diagrams		
4.0	Mechanical Checks		
4.1	Ensuring all required codes, standards and regulations are followed based on requirements		
4.2	Cabinet Material Cert		
4.3	Checking for appropriate fit of panels, doors, hinges, hood, adjustment of legs, pipe supports & required pipe sleeves. Tightness of flared fittings, and that no leaks are found.		
4.4	Availability of components and their tags as per P&ID (list along with the purpose of following components to be provided by Vendor) - Tanks - Pumps - Heat Exchangers - Filters - Manual Valves - Auto valves - Check/Non-Return Valves and their direction - Pressure Regulators - Back pressure valves - Pressure Relief valves - Flowmeters - rotameters - Pressure transducer/transmitter - Pressure switches - Pressure gauges - Temperature sensor/transmitter - Conductivity sensor/transmitter - Densitometer - Concentration meter		

S/N	Items for Inspection	Verification	Satisfactory Remarks
	- Sensors for door status, leak, drain, vent, empty, valve status, discrete level, etc. - UV/IR flame detectors - Load cells - Cooling fan for elect cabinet		
4.5	Torque (for tightness) check has to be performed for required components. Random check shall be performed if too many to check.		
4.6	Leak test for the cabinet containment area and other relevant sections in the cabinet such as sampling box, etc.		
5.0	Tanks		
5.1	Ensuring all required codes, standards and regulations are followed based on requirements		
5.2	Manufactures certificate		
5.3	Tank sizes		
5.4	Bottom shape and requirement		
5.5	Installation of dip-tubes		
5.6	Installation of spray balls based on drawings		
5.7	Leak test		
6.0	Pumps		
6.1	Check Pump Models to correspond P&ID		
6.2	Check Inlet and Outlet sizes		
6.3	Availability of surge suppression system		
6.4	Check Availability of Cooling line		
7.0	Interfaces – Size and Ratings		
7.1	Chemical		
7.1	Chemical Drain Lines		
7.2	Instrument Air Interface Line		
7.3	CDA Interface Line		
7.4	Process and Utility N2 interface Lines		
7.5	UPW Interface Lines		
7.6	Process Cooling Water		
7.7	Interconnection Lines		
7.8	Exhaust Line		

S/N	Items for Inspection	Verification	Satisfactory Remarks
8.0	Electrical and Controls		
8.1	Ensuring all required codes, standards and regulations are followed based on requirements		
8.2	Availability of list of Components in Electrical Cabinet (list of following components to be provided by Vendor) - EMO - Tower light - ELCB/RCB and other circuit breakers - Terminal blocks - VFDs - Fuse blocks		
8.3	Availability of list of PLC Components in Electrical Cabinet (list of following components to be provided by Vendor) - PLC with Removable Memory card - HMI with Removable Memory card - Communication cards - Digital and Analog cards - Segregation of cards based on segregation of engines if necessary - Communication links through DLR and check for dual link communication. Check if activation is needed and completed.		
8.4	Availability of list of Components in Pneumatic Compartment - Solenoid blocks - Segregation of solenoid blocks based on individual engines		
8.5	Do electrical panels meet required IP standard or other code of practices – prove documents to be included in manual		
8.6	Check cable conduits, din rails, solenoid valve back panels, breakers, etc. are securely mounted. Make sure no sharp edges are present.		
8.7	Check lugs and ferrules are tight and according to specification		
8.8	Panel earth connections if required		
8.9	Location of components as per approved drawings		
8.10	Check Tagging and labelling are correct		
8.11	Earthing continuity check		
8.12	Wiring terminal check		
8.13	Check if all grounding is done correctly		

S/N	Items for Inspection	Verification	Satisfactory Remarks
8.14	Continuity of Safety Earth Bar (Power Earth – PE) and interconnecting links		
8.15	Continuity of Instrument Earth Bar (IE) and interconnecting links		
8.16	Ensure integration is done as per architecture drawing		
8.17	Dry I/O Test – All continuity to be tested and proved.		
9.0	Instrumentation		
9.1	Ensuring all required codes, standards and regulations are followed based on requirements		
9.2	Door and panel switches - List along with functions shall be provided by vendor		
9.3	Leak Sensors - List along with functions shall be provided by vendor		
9.4	Empty Sensors – List along with functions shall be provided by vendor		
9.5	Level Sensors – List along with functions shall be provided by vendor		
9.6	Flow meters - List along with functions shall be provided by vendor		
9.7	Pressure switches - List along with functions shall be provided by vendor		
9.8	Pressure Transduces - List along with functions shall be provided by vendor		
9.9	Temperature Sensors - List along with functions shall be provided by vendor		
9.10	Conductivity Sensor/Transmitter - List along with functions shall be provided by vendor		
9.11	Density Meter - List along with functions shall be provided by vendor		
9.12	Load Cells - List along with functions shall be provided by vendor		
9.13	Pump leak sensors for pneumatic pumps - List along with functions shall be provided by vendor		
9.14	Z-purge instrument based on necessity - functions shall be provided by vendor		
9.15	RPM – Counters if any		
9.16	Thermostats		

S/N	Items for Inspection	Verification	Satisfactory Remarks
10.0	System Power Supply Check		
10.1	Ensuring all required codes, standards and regulations are followed based on requirements		
10.2	Verify all cabinet circuit breakers are open/off		
10.3	Verify all knife edge field terminals are open		
10.4	Connect the required power supply from field to incoming terminals		
10.5	Before switching on MCBs, ensure no voltage is present at the reference point		
10.6	Before switching on MCBs, ensure the system is ready for power supply. Any loose connections or any connections that can cause problems to be rectified.		
10.7	Operate Distribution MCBs within the cabinet and confirm correct voltage		
	Specified Voltage A/C: Actual Measure Voltage:		
	Specified Voltage D/C: Actual Measure Voltage:		
11.0	Wet I/O Testing		
11.1	All pneumatic operated valves shall be tested by manual activation at solenoid side and HMI side		
11.2	All door and panel switches shall be tested by vendor		
11.3	All leak Sensors with functions shall be tested by vendor		
11.4	Empty Sensors – List along with functions shall be tested		
11.5	Level Sensors – All functions (low, high, filling, stop, etc.) shall be tested		
11.6	Flow meters - All functions (low, high, etc.) shall be tested		
11.7	Pressure switches - All functions (low, high, etc.) shall be tested		
11.8	Pressure Transduces - All functions (low, high, etc.) shall be tested		
11.9	Temperature Sensors - All functions (low, high, etc.) shall be tested		
11.10	Conductivity Sensor/Transmitter – All functions shall be tested		
11.11	Density Meter - All functions shall be tested		

S/N	Items for Inspection	Verification	Satisfactory Remarks
11.12	Pump leak sensors for pneumatic pumps - All functions shall be tested		
11.13	Z-purge instrument based on necessity – All functions shall be tested		
11.14	RPM – Counters if any		
11.15	Thermostats and its functions to be tested		
11.16	Load Cells - All functions shall be tested		
11.17	Load cell taring and repeatability		
11.18	Test Related to Pumps		
11.19	VFD settings and ramp-up and ramp down to be tested through mA injection or some other means		
11.20	Check VFD digital signals and their appropriation		
11.21	Max RPM setting based on necessity		
11.22	Whatever field instruments cannot be tested due to unavailability of instrument, FAT testing shall be performed by injecting appropriate 24V DC or required mA.		
11.23	NPSH testing when max flow takes place – If system is hooked up to a loop only this test can be performed.		
12.0	Functional Test		
13.0	Alarm list and testing of alarm list – Vendor needs to provide the alarm list and needs to create such alarms		
14.0	Scenario Tests – Vendor needs to provide list of all possible different scenarios such as tank switch, preparation, qualification, engine failure, auto filter change, etc. and actual sequence of operation during these scenarios.		
15.0	Pressure control modes and appropriate tests		

Note:

1. The above table is only a guideline and actual full list shall be provided by the vendor.

Appendix 2 - Guideline for Installation Inspection

Name of Inspector:	FAT Date:
Name of Organization:	Person in Charge from Vendor:
Name of Fab:	Name of Vendor/Supplier:
Name of System:	Job/PO Number:

S/N	Items for Inspection	Verification	Satisfactory Remarks
1.0	Readiness of vendor with Installation Inspection checklist		
2.0	Availability of approved P&IDs and Lay Out Drawings		
3.0	Availability of single line diagrams and Electrical & I/O Panel Installation Lay Outs		
4.0	Mechanical Checks		
4.1	Ensuring all required codes, standards and regulations are followed based on requirements		
4.2	Ensuring the installations is as per approved lay out drawings and all the installations are fastened properly, securely and firmly on mounting supports.		
	- Day Tanks		
	- Chemical Transfer Units		
	- Chemical Blending Units		
	- Chemical Dispense Units		
	- Chemical VMBs		
	- Chemical Sump Pump Systems		
4.3	Ensure all weld tags according to Semi S2 Standard		
	Ensure the installation integrity of the major components such as pumps, filters, valves, regulators, pipes, etc. – Individual Checklists are to be submitted		
	Ensure the ratings of pumps and filters as per requirement – Individual Checklists are to be submitted		
	Ensure the mechanical installation integrity of the major instruments such as flowmeters, conductivity meters, concentration meters, pressure transmitters, etc. – Individual Checklists are to be submitted		
4.4	Ensure all Labelling and component Tags According to Semi S2 Standard		
4.5	Ensure all edges of Unistrut are filed and covered with approved end covers to avoid any injuries		
4.6	Ensure usage of appropriate chemical resistant anchor bolts and anchoring accessories		
4.7	Ensure Appropriate leveling legs used for the system		
4.8	Ensure Levelling of the system done appropriately such that the system does not exert any pulling or pushing effect to the associated pipes and supports		
4.9	Ensure appropriateness of Drain Line design		
4.10	Ensure appropriateness of drain lines connections		
4.11	Ensure Drain ports at relevant locations and the use of appropriate valves and plugs.		
4.12	Ensure there are no conflicts with other installations.		

S/N	Items for Inspection	Verification	Satisfactory Remarks
4.2	Pressure Ratings of Pipes and Tubes used in Chemical Rooms		
	- Chemical Tubes		
	- Chemical outer Containment Tubes		
	- Chemical Drain Lines		
	- Instrument Air Interface Line		
	- CDA Interface Line		
	- Process and Utility N2 interface Lines		
	- UPW Interface Lines		
	- Process Cooling Water		
	- Interconnection Lines		
4.3	Pressure Ratings of Pipes, Tubes and other components used at the Field (VMBs or any other remote chemical related units)		
	- Chemical Tubes		
	- Chemical outer Containment Tubes		
	- Instrument Air Interface Line		
	- CDA Interface Line		
	- Process and Utility N2 interface Lines		
	- UPW Interface Lines		
	- Process Cooling Water		
	- Interconnection Lines		
4.4	Material and Brand of Pipes and Tubes used		
	- Chemical Tubes		
	- Chemical outer Containment Tubes		
	- Chemical Drain Lines		
	- Instrument Air Interface Line		
	- CDA Interface Line		
	- Process and Utility N2 interface Lines		
	- UPW Interface Lines		
	- Process Cooling Water		
	- Interconnection Lines		
	- Exhaust Line		
4.5	Availability of Pressure Test Reports		
	Chemical Loop		
	Chemical outer Containment		
	- Instrument Air Interface Line		
	- CDA Interface Line		
	- Process and Utility N2 interface Lines		
	- UPW Interface Lines		
	- Interconnection Lines between Day Tank and System		
	Interconnections between Transfer unit and Day Tank		
	Interconnections between Blender Unit and Day Tank		
	Interconnections between Bulk Tank to other transfer/blending/dispense sections		
	Tanks		
	VMBs if it is required		

S/N	Items for Inspection	Verification	Satisfactory Remarks
4.6	Material and Brand of other Installation Components		
	Exhaust Dust		
	Exhaust Damper		
	Exhaust Pressure indicator/transmitter		
	Solenoid block for VMB		
	VMB Valves		
	VBM Leak Sensors		
	Pipe Support Hangers		
	Pipe Support Clamps		
	Pipe Support Unistrut		
	Pipe Support		
4.7	Sump Pump System and its components		
	Pump Cabinet		
	Sump pit and sump tank checks for its integrity		
	Capacity of pump for the required usage		
	Sizes of Drain Line Connections to Sump Pit and Sump Tank		
	Capacity of Drain Line Pipes		
	Integrity of Piping system		
	Requirement of double containment		
5.0	- Electrical Checks		
5.1	- Follow Required Installation Inspection Checklist specified in Electrical Specification		
5.2	Ensuring all required codes, standards and regulations are followed based on requirements		
5.3	Ensuring the installations is as per approved lay out drawings and all the installations are fastened properly, securely and firmly on mounting supports.		
5.4	Ensure Control cables are not bundled with power cable to avoid any EMI related issues.		
5.5	The panel passes a visual inspection and complies with the following:		
	Ensure Outer enclosure of Electrical, I/O Panel, and Electrical & Control panel of the system are labeled appropriately with ID tag.		
	Ensure Devices and modules are installed per general assembly drawings and single line diagrams.		
	- Wiring, devices and modules are labeled within the panel in accordance with the Single Line and general assembly drawings.		
	- Wiring terminations (with lugs, ferrules, etc.) are adequate and firm, and wiring utilizes the color code established within the drawings.		
	- Wiring cable shields are terminated at one end to power supply ground as appropriate.		
	- Wiring is gathered with tie-wraps or bundled within wire-mold channels.		
5.6	- Verify Ratings of all cabinet circuit breakers		
5.7	- Verify all incoming and outgoing connections to and from the panels. Other ends need to be verified.		

S/N	Items for Inspection	Verification	Satisfactory Remarks
5.8	Ensure all the electrical and control connections to I/Os and controllers of Pumps, VFDs and associated controllers – Individual Checklists are to be submitted		
5.9	Ensure all the electrical and control connections to I/Os and controllers of instruments, transmitters, etc. – Individual Checklists are to be submitted		

Note:

1. When chemical is mentioned in the installation inspection, it is to be noted that both concentrated chemical and blended chemical are considered as chemical.
2. As components in the system are checked during FAT and SAT, the verification of the system components is not necessary unless there is any change done after FAT or SAT.
3. If the components are more than one, the vendor shall prepare the list based on the actual installation data.
4. Installations related to Fire Sprinkler or other Safety Related Installations will have to be se be inspected by relevant parties.

Appendix 3 – Chemical Receiving Checklist

Chemical Receiving Checklist			
Chemical System Vendor		Date/Time of Delivery	
Chemical Supplier		Supervisor in Charge	
Area of Inspection		Inspected By	
S/N	Items	Yes/ No/ NA	Actions / Comments
1	Prior to Delivery		
1.1	Confirm chemical name, container type, and quantity to be delivered onsite.		
1.2	Is calcium gluconate onsite for hydrofluoric acid delivery?		
1.3	Confirm hazard type for chemical to be delivered and any specific safety requirements.		
1.4	Ensure access is prepared - area clear with no materials or vehicle obstructions		
1.5	Ensure all safety paths are cleared for access to safety shower and for evacuation		
1.6	Ensure safety showers are tested and functional		
1.7	Are all room life safety devices tested and functional - fire suppression, fire sprinklers, fire alarms, etc.		
1.8	Ensure room exhaust is active and adequate		
1.9	Ensure all PPE, spill kits, fire extinguishers, SDS, and first aid kits are present		
1.10	Ensure proper chemical signage, including room signage is displayed and the appropriate locations		
2	Onsite Delivery / Receiving of Chemicals		
2.1	Is the point of transferring being guarded and kept out of unauthorized personnel and work?		
2.2	Are proper signs erected to caution/warn unauthorized team members to keep out?		
2.3	Are other incompatible works being asked to cease during the transferring of chemical - like hot work, grinding, etc.		
2.4	Are incoming chemical containers in good condition		
2.5	Verify containers are labeled correctly and SDS is present		
2.6	Check that all delivery crews are wearing appropriate PPE		
3	Storage of Chemicals		
3.1	Is storage space sufficient for qty of chemical to be delivered		
3.2	Sufficient passageway and access are provided to store and deliver chemical for use		
3.3	Storage racks of adequate strength with labels for storage weight limits and chemical types		
3.4	Chemical bottle/carboy labels face outward for ease of identification and retrieval		
3.5	Incompatible chemicals are isolated and stored separately		
3.6	Adequate ventilation and safety equipment are provided and readily available		
4	Handling and use of Chemicals		
4.1	Workers know the chemical name and safe handling procedures		
4.2	Workers have undergone hazard communication by their leader		
4.3	Workers have access to SDS at area of work at all times		
4.4	Workers are wearing the appropriate PPE		
4.5	Workers are aware of procedure for spill clean up and location of spill kits		
4.6	Appropriate waste collection bins are in place and labeled		
4.7	Workers know proper disposal procedures		
4.8	Workers are aware of location of nearest emergency shower/eyewash station in case of chemical exposure		
4.9	Workers observe personal hygiene after handling of chemicals		

Appendix 4 – Checklist for First Charge of Chemical

Preparation Checklist for first Charge of Chemical into System & Loop Distribution			
Inspected By		Date/Time	
Area of Work		Chemical Name	
Equipment ID		Chemical Category	
S/N	Items	Yes/ No/ NA	Remarks
1	Have all piping and drain lines passed pressure testing?		
2	Have all chemical system leak sensors, interlocks, pressure relief devices and EMO been tested in AUTO mode? Has Micron verified?		
3	Are all valves in chemical and drain piping in closed and capped position?		
4	Are all utilities systems turned ON and in required spec with stable supply ? - Power - Exhaust (for chemical system, room emergency exhaust) - UPW - CDA - N2 - Instrument Air (if applicable) - City water (for Emergency shower)		
5	Are all containment boxes closed and sealed properly?		
6	Are all distribution valves manifold boxes closed and sealed properly?		
7	Have all chemical distribution VMB/MHB/MHEB leak detects been tested with the chemical system in AUTO and has Micron verified that the chemical system shuts down when those leak detects are activated?		
8	Has the sump been tested and in AUTO mode?		
9	Are all exhaust systems turned ON and balanced - chemical system, sump, and room?		
10	Is Solvent system (tank, piping & VMB) connected to appropriate grounding bar?		
11	Are the waste systems ready and available?		
12	Has the quantity and timing of chemical being flushed to WWTP and waste chemical being trucked out to outside vendor been verified and communicated to the WW Operations team?		
13	Is the SDS readily available and location known?		
14	Has ERT been informed of the charging of chemical and on standby for the event.		
15	Is the ERT contact information posted in the chemical room?		
16	Has the charging of chemical been communicated to the ENTIRE work force in the affected building?		
17	Are there cautionary signs in the area of chemical charging, including all areas of the sub-fab where the chemical piping is present?		
18	Are spill kits, first aid box, and PPE in the area of work?		
19	Are the emergency eye wash/shower stations tested and ready, including all areas of the sub-fab where the chemical piping is present?		
20	Are all life safety systems - fire sprinklers, fire alarms & fire suppression system (where applicable), are tested and ready in the chemical room, including all areas of the sub-fab where the chemical piping is present?		

Appendix 5 – Utilities Interfaces and Requirements

UTILITIES / CATEGORY		Chemical Room	Sump Pump System	Chemical System
Electrical		- Room EMO - Room Lighting and power sockets - Power supply to Room electrical panels - Earth Grounding bar	Power supply to sump system	Power supply to system
I&C		- Room EMO at door exits - Room door card access	Sump system alarms to SCADA / FMCS.	- Individual System EMO and Global EMO at FMCS or TGCM room. - All required system alarms to SCADA. - Required instrumentation information and interlocks.
Mechanical	IA or CDA (if IA is not available)	20% spare points	Supply of required amount of CDA for sump pumps.	Supply of required amount of IA or CDA for valves & other instrumentation.
	Emergency Exhaust	Calamity Exhaust - activation in the event of large spill.	NA	NA
	HVAC	- Room Temperature shall be maintained at 20-25 deg C. - Relative humidity < 65 % - Sufficient air exchange as per room hazard classification and individual system air exchange requirement.	NA	NA
	PCW	20% spare points - Based on requirement, Chilled water may be used for Heat Exchanger.	NA	Depending on system requirement
	Acid / Caustic / Solvent Exhaust	20% spare points - For some chemical rooms the make-up air room main outlet and exhaust inlet ducts have to be installed down to 50cm above the floor (to be determined based on chemical properties).	Required for Sump pump cabinet and sump tank.	- Meet code requirement if any. - Depending on the nature of the chemical.
	Door Air Curtain	Depending on high humidity regions, an automatic air curtain shall be installed on the outer door of chemical rooms where drum/tote delivery frequently happens.	NA	NA

UTILITIES / CATEGORY		Chemical Room	Sump Pump System	Chemical System
	Emergency Eye Shower	As per EHS requirement	NA	NA
	Fire Sprinkler	As per relevant Fire code requirement.	NA	Not required unless otherwise specified.
Gas	GN2	20% spare points	NA	1. Blanketing 2. Purging
	PN2	20% spare points	NA	Based on requirement GN2/ UN2/ PN2 will be used.
Water / Waste	UPW	1. 20% spare points 2. UPW Return Line Valve Manifold assembly (except for solvent room)	NA	1. Preparation / Recipe 2. Flushing & Rinsing 3. Humidification
	Drain / Waste sump	Sump tank / pit connection to appropriate sump system.	Double contained connection to appropriate drain header.	Drain connection to sump tank.
	Non-Portable / City Water	For Emergency Shower & Eye Wash units	Drain line to sump tank	NA
Civil		1. Room Doors with Push Bar Safety Panic Bar Exit handle 2. Sump Pit(s) 3. Equipment Move-in access 4. Trenches with sufficient gradient leading to sump pit. 5. Provisions for pipe supports 6. Loading bay and Drum transport passage. 7. Discharged Sprinkler water gravity drainpipe to designated collection tank. 8. Tank plinths and bund wall (if required during design phase) 9. Removable Wall for bulk tank access and blast relief wall for solvent rooms	1. Sump Pit 2. Trenches leading to sump pit	
EHS		LSS Alarms Room PPE Cabinets	LSS Alarms	

Note: Chemical System refers to chemical dispense, transfer, blender and other systems containing chemical.

Appendix 6 - Documents to be submitted at different phases of the project by vendor

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
1	Company Information and corporate overview with Local Organization Chart	Progressive report upon Changes	Progressive report upon Changes	Progressive report upon Changes	Progressive report upon Changes	
2	Past projects or References	NA	NA	NA	NA	NA
3	Standard Project Management and Project Control Flowchart	Project Management and Project Control Flowchart relevant to this project	Accepted Project Management and Project Control Flowchart	Progressive report upon Changes	Progressive report upon Changes	Applied Project Management and Project Control Flowchart
4	Standard Project Organization Chart	Project Org Chart with names	Progressive Project Org Chart with names upon changes made – Installation team needs to be included with details.	Progressive Project Org Chart with names upon changes made – commissioning team needs to be included with details.	NA	Final Project Org Chart and Contact List of key personals
5	Relevant approvals and certificates for manufacturing slurry/chemical systems and Tanks	Manufacturing site information and required certificates including Clean Room (Clean Environment) Certificates (or Information) for system manufacturing	Manufacturing site information and required certificates and acceptance from Micron	NA	NA	Relevant approvals and certificates for manufacturing slurry/chemical systems and Tanks

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
6	Scope work Matrix	Revised scope matrix without deviating the agreed scope as per PO	Progressive scope matrix upon agreed changes	Progressive scope matrix upon agreed changes	Progressive scope matrix upon agreed changes	Executed Scope Matrix and Gaps between agreement and execution.
7	Schedule with Critical Path	Schedule with Critical Path for approval with actual dates honoring the agreement as per PO	Progressive schedule with appropriate mitigation schedule.	Progressive schedule with appropriate mitigation schedule	Progressive schedule with agreed completion dates	
8	Standard Process Flow Diagrams	Process Flow Diagrams	Reviewed PFDs	Approved PFDs	NA	As-Built PFDs
9	Standard P&IDs	System P&IDs	Reviewed System P&IDs	Approved system P&IDs	NA	As-Built system P&IDs
10	Layout Drawings for different rooms	Layouts and Initial Solid works or equivalent that can be used for BIM Development systems piping, electrical and control networks	Reviewed Layouts and stages of BIM showing all conflicts systems piping, electrical and control networks	Approved Layouts and stages of BIM with existing conflicts systems piping, electrical and control networks	NA	Soft copy of As built interactive (appropriate for future use) layouts and BIM softcopies.

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
11	Standard System Design Features	System Design Features	Reviewed design features and Shop Drawings before starting fabrication	Approved Shop Drawings before starting fabrication	NA	As built system drawings
12	Technical proposal that include: Design Basis and calculations Performance guarantee Description of major processes Equipment supply and specifications Piping specifications Utility Interfaces and requirements	Technical information with respect to design	Reviewed technical information	Approved technical information.	NA	Final Technical Information that include: Design Basis and calculations Description of major processes Equipment supply and specifications Piping specifications Utility Interfaces and actual utility consumptions Final Performance and gaps

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
13	Standard Component List submittals	Component list submittals: - Equipment list - Instrumentation list - Mechanical Components list - Electrical Panel/ Component list - PLC/ SCADA/ Network related Hardware/ Software Components list - I/O list	Reviewed component list submittals: - Equipment list - Instrumentation list - Mechanical Components list - Electrical Panel/ Component list - PLC/ SCADA/ Network related Hardware/ Software Components list - I/O list	Approved component list submittals: - Equipment list - Instrumentation list - Mechanical Components list - Electrical Panel/ Component list - PLC/ SCADA/ Network related Hardware/ Software Components list - I/O list	NA	Final Component List - Equipment list - Instrumentation list - Mechanical Components list - Electrical Panel/ Component list - PLC/ SCADA/ Network related Hardware/ Software Components list - I/O list
14	Software that will be used for development of the system – CFD, AFT Fathom or PIPEFLO, 3D Modelling, BIM, etc.	Software that are used for design calculations and for development of the system – CFD, AFT Fathom or PIPEFLO, 3D Modelling, etc.	- Computational Fluid Dynamics (CFD) results for required processes. - Results of hydraulic modeling based on AFT Fathom or PIPEFLO - 3D Modelling for system manufacturing	Approved - Computational Fluid Dynamics (CFD) results for required processes. - Results of hydraulic modeling based on AFT Fathom or PIPEFLO - 3D Modelling for system manufacturing	NA	Soft copy of As built and interactive (appropriate for future use) modelling information of CFD, AFT Fathom or PIPEFLO, 3D Modelling for systems, piping, etc.

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
15	Planned Control Network Architecture	Control Network Architecture	Control Network Architecture with - Graphic, Loop check documents - I/O loop check list - Instrument – HMI - SCADA loop test with alarm and failure test list - Instrument - HMI - SCADA test forms	Approved Control Network Architecture and related documents. - Graphic, Loop check documents - I/O loop check list - Instrument – HMI - SCADA loop test with alarm and failure test list Instrument - HMI - SCADA test forms	Progressive check lists	As-built Control Network Architecture with signed related documents - Graphic, Loop check documents - I/O loop check list - Instrument – HMI - SCADA loop test with alarm and failure test list Instrument - HMI - SCADA test forms
16	Electrical plans	General arrangement electrical drawings	Electrical drawings, I/O panel drawings and single line diagrams	Approved electrical & I/O panel drawings and single line diagrams	Progressive check lists	As-Built electrical drawings, electrical panel layout drawings, I/O panel drawings, single line diagrams.
17	Standard Sequence of Operation	Sequence of Operation – Theory of Operation shall be included	Reviewed SOO - Theory of Operation shall be included. Functional test with interlocks and scenarios shall be included	Approved SOO - Theory of Operation shall be included. Approved Functional test with interlocks and scenarios shall be included	Progressive check lists	Final SOO with the with signed functional test reports.

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
18	Standard VMB and MHB/MHEB drawings	VMB and MHB/MHEB Drawings		Approved VMB and MHB/MHEB Drawings	Progressive check lists	As-Built VMB and MHB/MHEB Layout Drawing
19	Standard HAZOP and FMEA	HAZOP and FMEA – This should lead safe and consistent system operation	HAZOP and FMEA outcome and report for necessary actions	Verification checklist if HAZOP and FMEA conclusions are followed in the systems and installations	HAZOP and FMEA related Approved check list for verification of systems and installations	Final FMEA and HAZOP analysis reports
20	Standard installation guidelines	Reviewed installation guidelines suitable for chem/slurry systems.	Approved installation guidelines suitable for chem/slurry systems.	Progressive check lists	NA	Final installation guidelines used during installations.
21	Standard commissioning plan	Commissioning plan	Reviewed commissioning plan	Progressive commissioning plan with change of any schedule and mitigation plan.	Progressive commissioning plan with change of any schedule and mitigation plan for completion.	Completion Certificates
22	Standard qualification procedure	Qualification procedure for systems and other installations	Reviewed qualification procedure for systems and other installations	Approved qualification procedure for systems and other installations	Progressive check lists	Signed copy of actual usage of approved qualification procedures.
23	Standard alarm list with alarm category	Alarm list and alarm category with reasons for any particular alarm.	Reviewed alarm List and alarm category with reasons for any particular alarm.	Approved alarm list and alarm category with reasons for any particular alarm.	Progressive check lists	Final alarm list with alarm category and priority
24	Standard pressure test procedure	Applicable pressure test procedures	Reviewed pressure test procedures	Approved pressure test procedures	Progressive pressure test reports	Final pressure test reports with relevant signatures

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
25	Standard FAT Information	FAT procedure and check list	Reviewed FAT procedure and check list with date for FAT	Approved FAT procedure and Checklist shall be used for FAT.	NA	Signed FAT Check lists.
26	Standard SAT Information	SAT procedure and check list	Reviewed SAT procedure and check list with date for SAT	Approved SAT procedure and Checklist.	Progressive SAT check lists	Signed SAT Check lists.
27	Standard installation inspection information	Installation inspection procedure and check list	Reviewed installation inspection procedure and check list	Approved installation inspection procedure and checklist.	Progressive installation inspection reports. Field service engineer's report on installation compliance	Signed installation inspection check lists.
28	Standard Leach Test Procedure	Applicable Leach Test Procedure – Based on chemistry	Reviewed Leach Test Procedure – Based on chemistry	Approved Leach Test Procedure – Based on chemistry	Progressive leach test reports	Leach test reports
29	Standard sampling procedure analytical methods	Applicable sampling procedure analytical methods	Reviewed sampling procedure analytical methods	Approved Sampling procedure and analytical methods used	Progressive analytical reports	Results of samples tested to be incorporated in commissioning reports
30	All other standard testing and commissioning information	All other testing and commissioning procedures and check lists	All other reviewed testing and commissioning procedures and check lists	All other approved testing and commissioning procedures and check lists including Functional test with interlocks and scenarios shall be included	Progressive related testing and commissioning reports.	All other signed testing and commissioning check lists and commissioning report

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
31	Standard calibration, verification and mapping information	Calibration, verification and mapping procedures and other information for the systems	Reviewed calibration, verification and mapping procedures and other information for the systems	Approved calibration, verification and mapping procedures and other information for the systems	Progressive reports and certificates	Final calibration, verification and mapping procedures and certificates
32	Standard analytical and physical parameters of the systems	Analytical and physical parameters of the systems to be critically monitored	Reviewed analytical and physical parameters of the systems to be critically monitored	Approved analytical and physical parameters of the systems to be critically monitored	Progressive reports on SCADA trends	All analytical and physical parameter test reports with SCADA trends for approval
33	Standard training program that includes schedule, categories of training and appropriate training contents	List of trainings to be provided for different categories	Reviewed list of trainings to be provided for different categories	Approved list of trainings to be provided for different categories	Progressive Training information	Training certificates and attendance List
34	Standard required test reports for tanks	Required test reports for tanks for appropriate chemistries	Reviewed required test reports for tanks for appropriate chemistries	Approved required test reports and PE certifications (based on regulatory requirements) for tanks for appropriate chemistries	Progressive certificates and reports	Final required test reports and PE certifications (based on regulatory requirements) for tanks for appropriate chemistries
35	Standard QA/QC plan and QA/QC procedures	Applicable QA/QC plan and QA/QC procedures	NA	NA	NA	NA

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
36	Standard required PE (or Regulatory Authority) certification for the required codes and regulations for systems, rooms and other related areas.	Applicable required PE (or Regulatory Authority) certification for the required codes and regulations for systems, rooms and other related areas.	Required PE (or Regulatory Authority) certification for the required codes and regulations for systems, rooms and other related areas.	PE (or Regulatory Authority) certification for the required codes and regulations for systems, rooms and other related areas before charging chemicals.	Progressive approvals and reports	Approval from PE or regulatory authority with relevant stamps and signatures.
37	Standard Change Control Forms	Applicable Change Control Forms	NA	NA	NA	Deviation List with approved deviation forms
38	Standard start-up and shut-down procedures	Applicable start-up and shut-down procedures	Reviewed start-up and shut-down procedures	Approved start-up and shut-down procedures		Final start-up and shut-down procedures
39	NA	NA	NA	NA	NA	Critical spare part list PLC, HMI and SCADA back-up programs PLC, HMI and SCADA licenses for the appropriate versions PLC, HMI and SCADA passwords for different access levels – Operator to Programming. Relevant on-site and off-site test reports and QA/QC documents

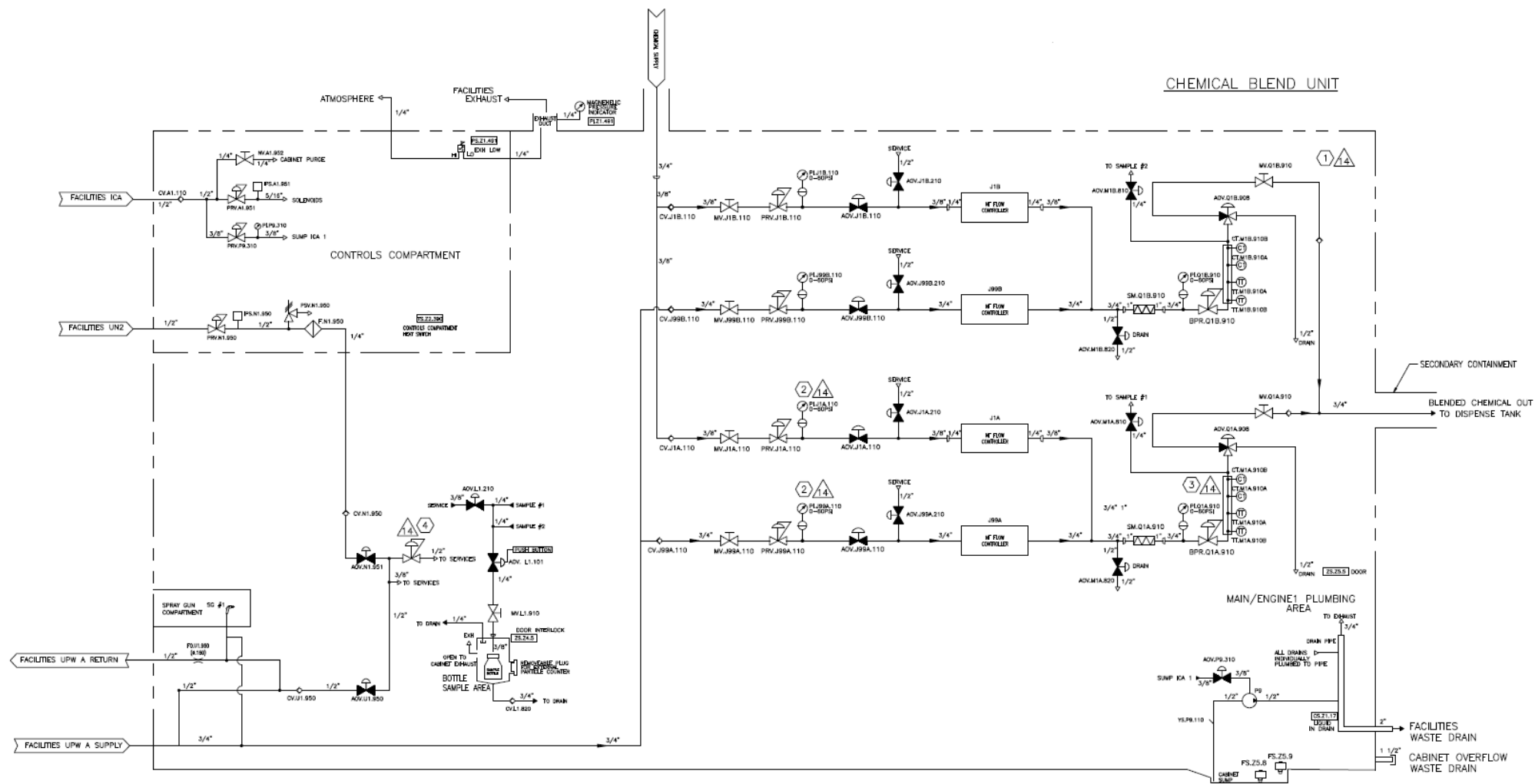
S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
40	Standard table of content for system operation and maintenance manual	Applicable table of content for system operation and maintenance manual	Reviewed table of content for system operation and maintenance manual	Approved table of content for system operation and maintenance manual	Final draft of O&M Manual as per approved table of content	System operation manual as per approved table of content
41	Submission of any other standard documents related to the project	Highlight any missing documents related codes, standards, regulations, etc.	Verification of any missing documents related codes, standards, regulations, etc.	Submissions of any missing documents related codes, standards, regulations, etc. to relevant authorities	NA	Approvals of any missing documents related codes, standards, regulations, etc.
42	Commercial proposal with required breakdown	Submission of any other Documents related to Issue For Construction (IFC)	NA	NA	NA	Approved completion certificate to start warranty of systems and components
43	Document submission plan	Document submission plan and applicable route.	NA	NA	NA	NA

Appendix 7 – Deviation Form

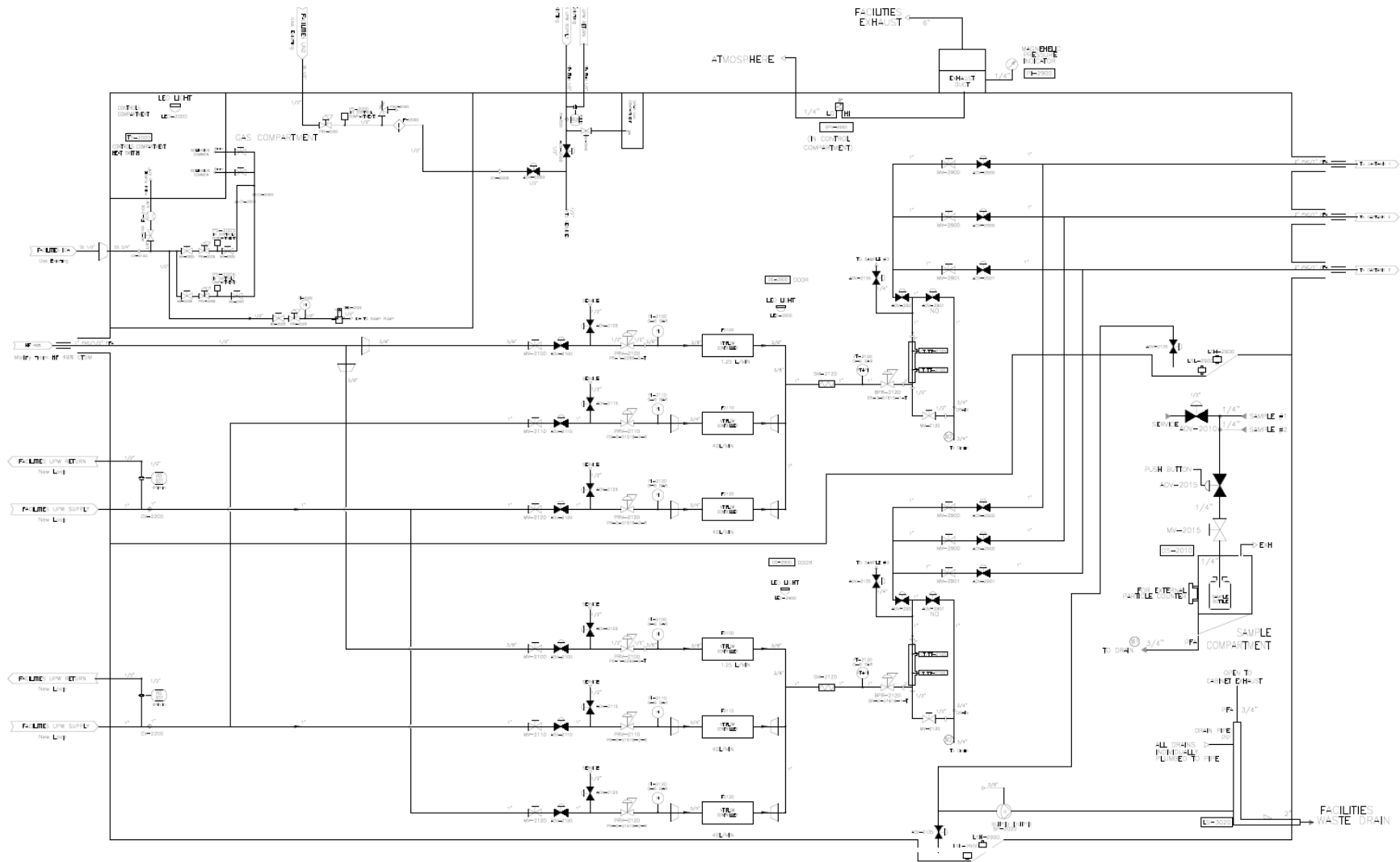
Site: _____		Name of the Standard Deviating from:	
Name of the Project: _____		Section(s) of the Standard Deviating from:	
Deviation Request Date: _____		Deviation Description:	
Name of Requester: _____			
Micron Dept./ External Vendor: _____			
Reason: (Technology / Material / Process / schedule / Cost / Space)			
Impacted Facilities Systems and Processes/ Workstations:			
Impacted Facilities System	Impacted Processes/ Workstations (if applicable)	Impact Description	
Justification Summary: (Includes links to data and backup slides as necessary)		Long term direction for permanent deviation:	
Mitigation plan for short term deviation:		Critical lessons learnt through this deviation if applicable:	
Approval:			
Department	Position	Name	Date
Site Facilities	Manager		
FCT	Manager		
GFTT	Manager		

Note: Any deviation from the given standard shall be submitted for approval. If a temporary deviation/exception is requested, the time needed for rectification shall be mentioned in the relevant box(es).

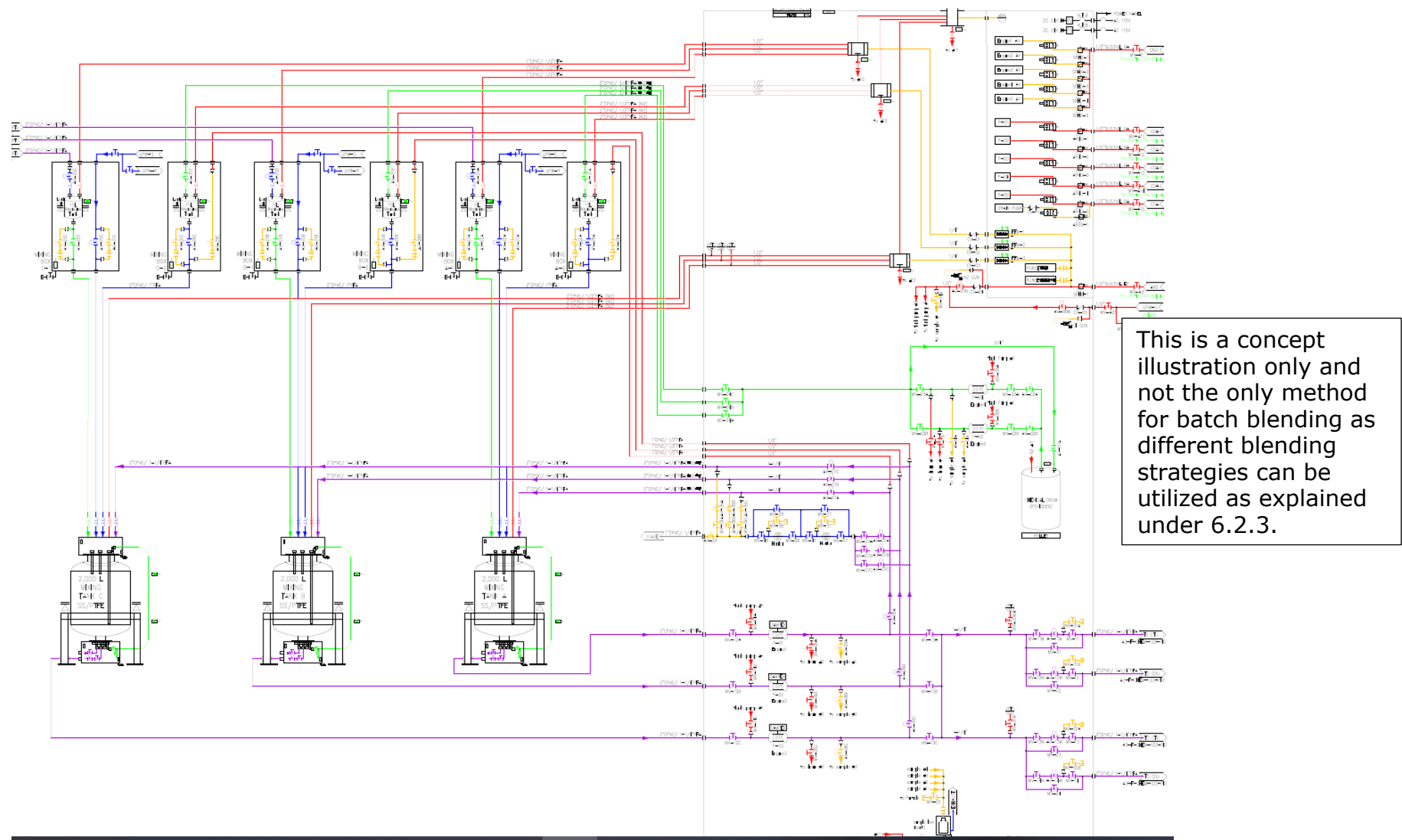
Appendix 8 – Typical P&ID: Chemical Blend Unit – Inline Type with Single Feed to Day tank.



Appendix 9 - Typical P&ID for Chemical Blend Unit – Inline Type with Multiple Feed to Day Tanks and with Additional UPW Flow Controller for High Capacity Blending. Similar Configuration can be Utilized for 3-Part Blending



Appendix 10 - Typical P&ID for Chemical Blend Unit – 3 Tank Batch Type (3-Part Blending)



Appendix 11

Typical P&ID for KP3100+STI2910 Batch blender. Blend Tank size and quantity needed for redundancy shall be determined during design stage.

